KMOU Division of Marine System Engineering

Introduction to Internal Combustion Engine

국립한국해양대학교
NATIONAL KOREA MARITIME & OCEAN UNIVERSITY

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Lesson content for each week

Ch0_Introduction to the Class

Ch1_Overview of the Internal Combustion Engine

Ch2_Thermodynamics and Compressed Air System

Ch3_Marine Fuels and Internal Fuel Oil System

Ch4_Intake and Exhaust System

Ch5_Turbocharger

Midterm

Ch6_Piston Fittings and Crankshaft

Ch7_Lube Oil System

Ch8_Cooling Water System

Ch9_Introduction to the Main Engine

Ch10_Alternative Fuels and Alternative Fuel Systems

Ch11_Environmental Regulations and Exhaust Gas Aftertreatment Systems

Ch12_Tools, Apprentice Engineer Duties, and Career Direction

Final Exam

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Tools

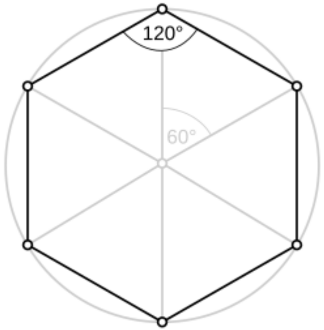
The place with the most tools in the engine room is the workshop.

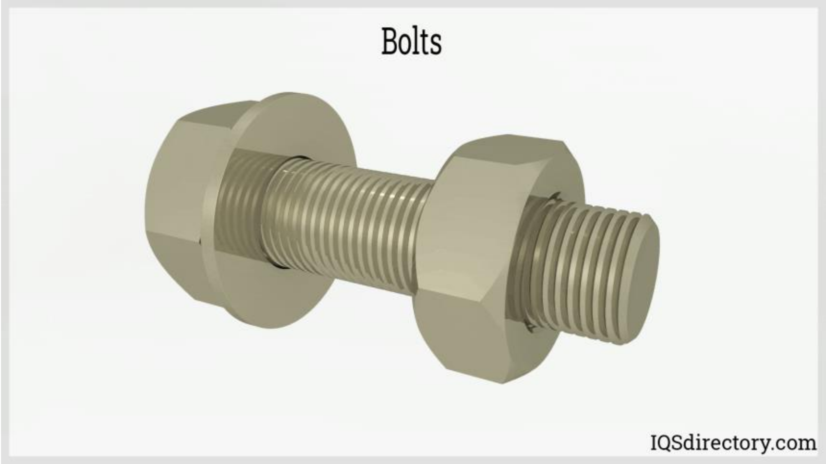
Some tools needed for work are also placed near the M/E and G/E.

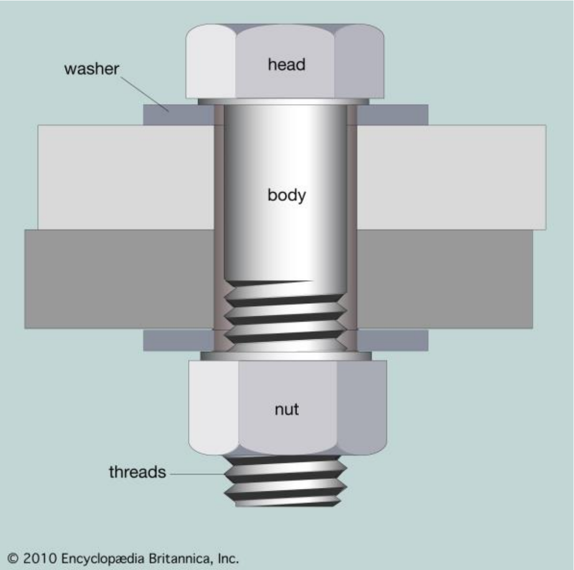
Bolt and nut

Hexagon bolt (= Hexagon head bolt, Hex bolt)

Hexagon



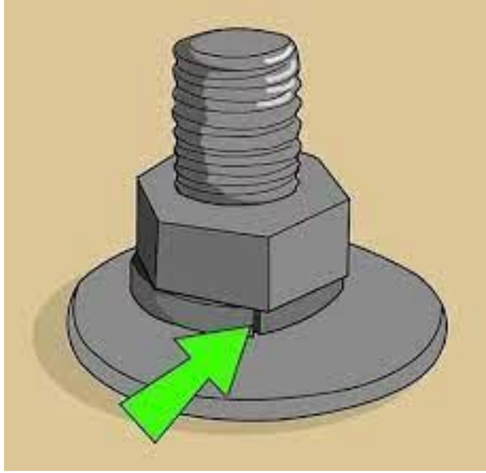


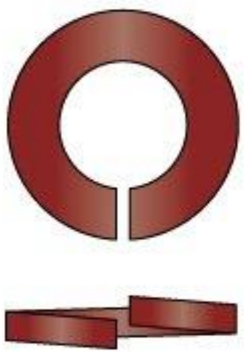


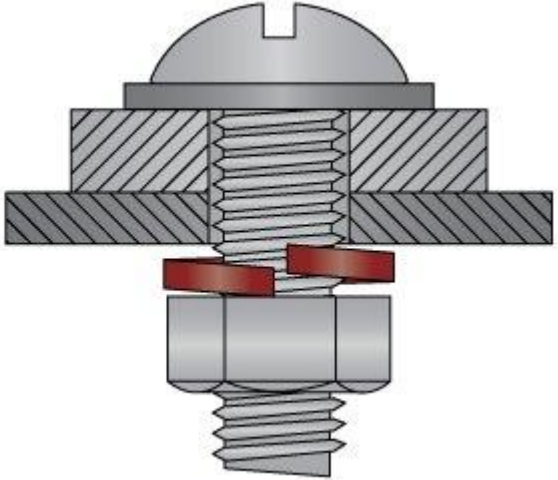
Washer

Plain washer (= Flat washer)

Spring washer





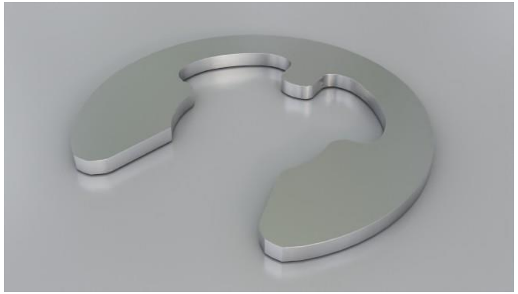


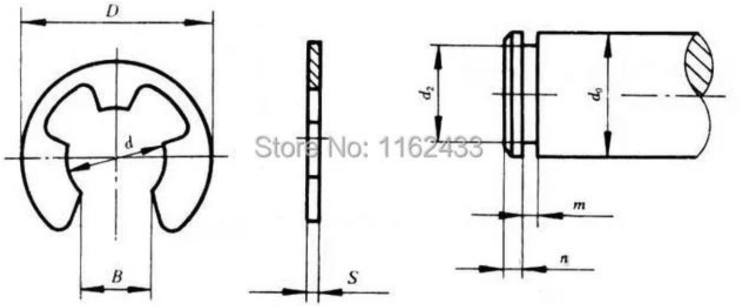
Circlip

Internal circlip

External circlip

E type circlip

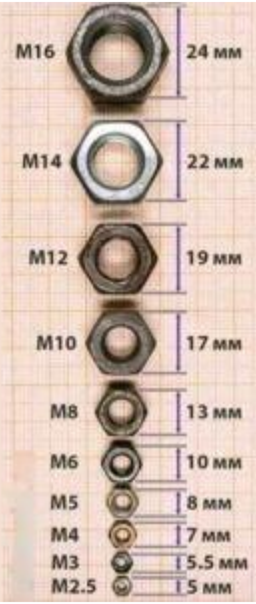




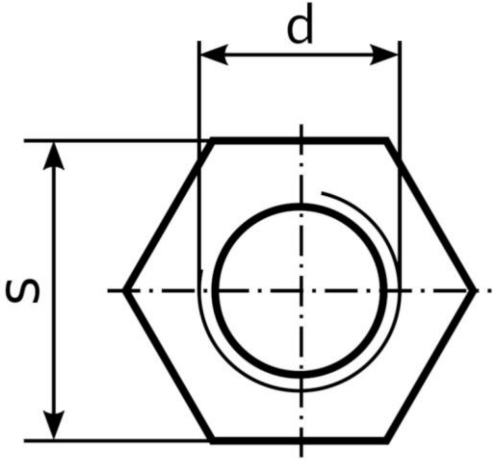
Tools

Single ended open spanner

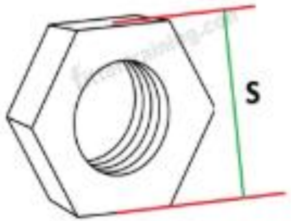
Tools

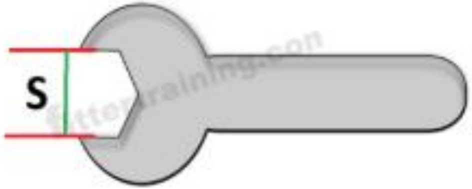


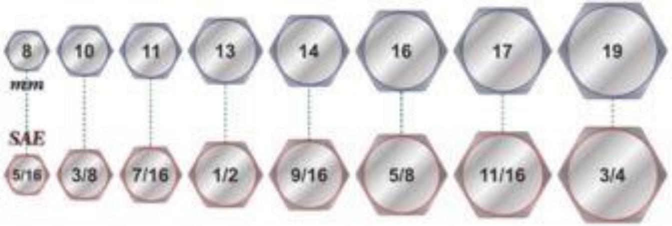
10mm = 1cm



d: nominal thread diameter
s: wrench size







Tools

Metric Spanner Size Chart for Bolts

Bolt Diameter (mm)	Spanner Size (mm)
M4	7
M5	8
M6	10
M8	13
M10	17
M12	19
M14	22
M16	24
M18	27
M20	30
M22	32
M24	36
M27	41
M30	46
M33	50
M36	55
M39	60

Tools

Double ended open spanner

Tools

Ring spanner

Tools

Combination spanner

Tools

Tools

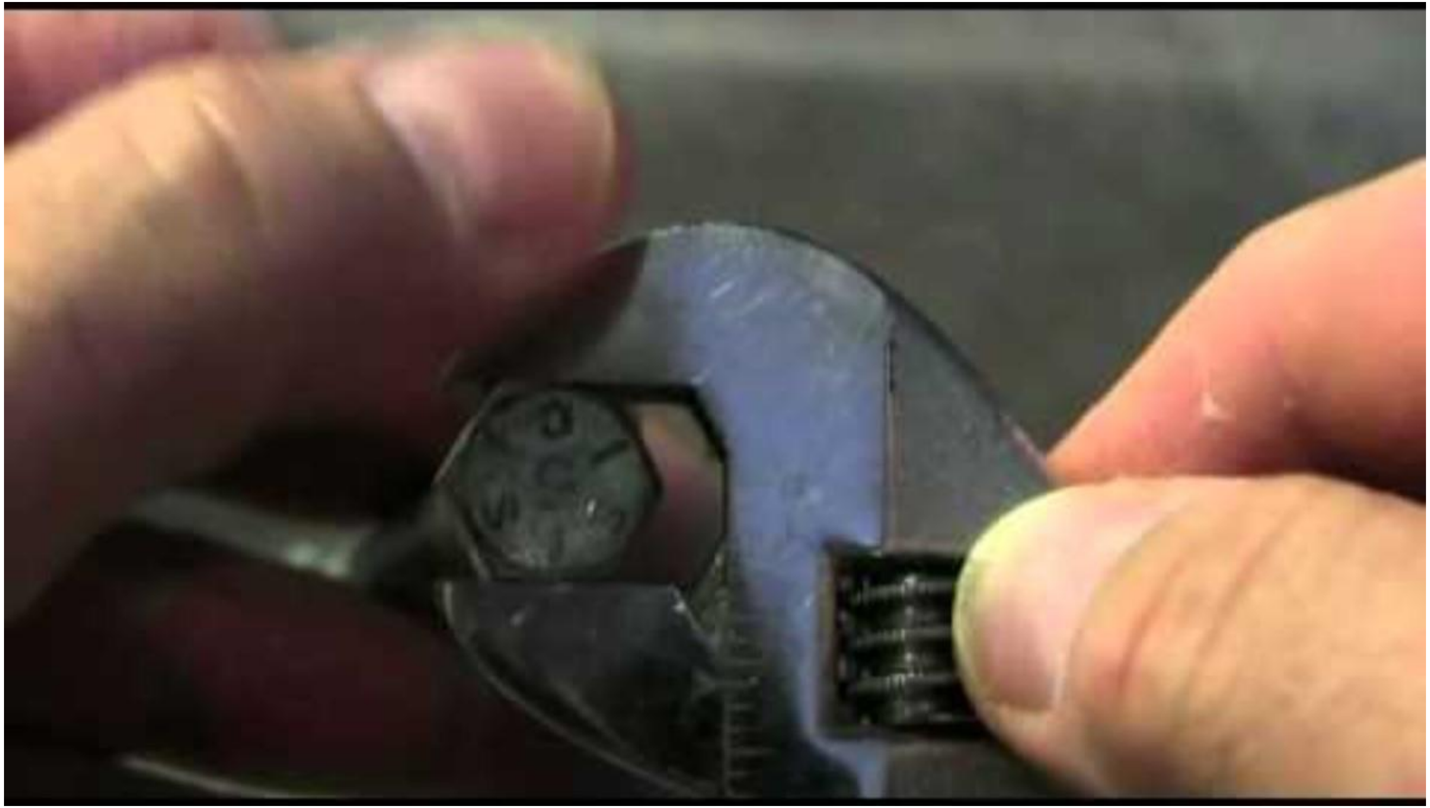

TOOL
TIPS TRICKS & HACKS
#1
OPEN ENDED
TECHNIQUE
TTG
TO THE
GARAGE

Bolt and nut

Tools

Adjustable spanner (=Monkey spanner)

Tools



Tools

Striking wrench (ring type)

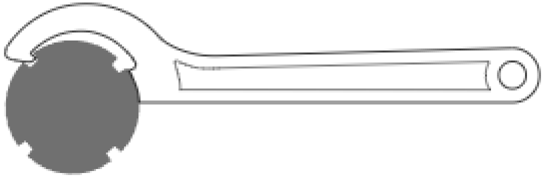
Tools

Striking wrench (open-end type)

Tools

Hook spanner

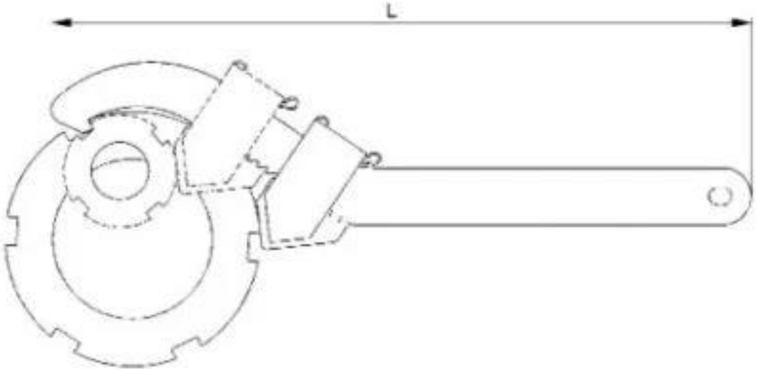
【How to use】



Hooks into the groove of a grooved round nut.

Tools

Adjustable hook spanner



Tools

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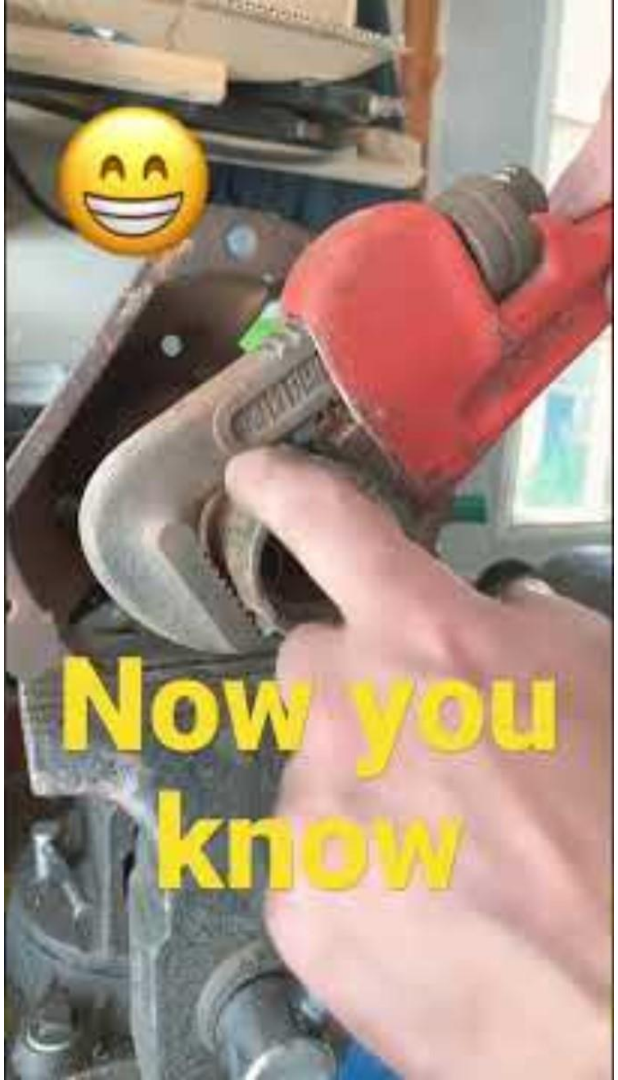
Tools

Pipe wrench

Tools



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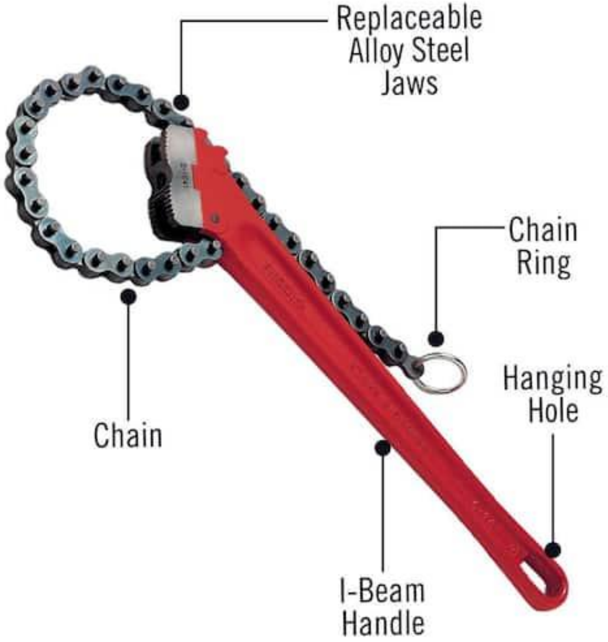
Tools

Tools

Chain pipe wrench



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Tools

Tools

Socket and

Ratchet wrench

Tools

Tools

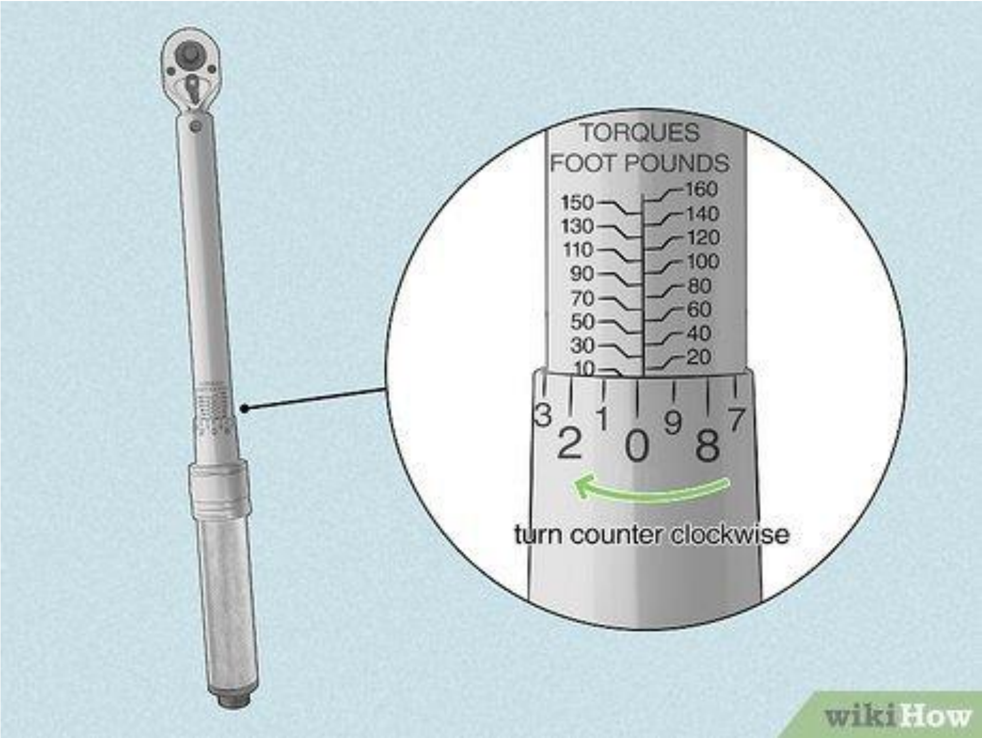
Tools

Tools

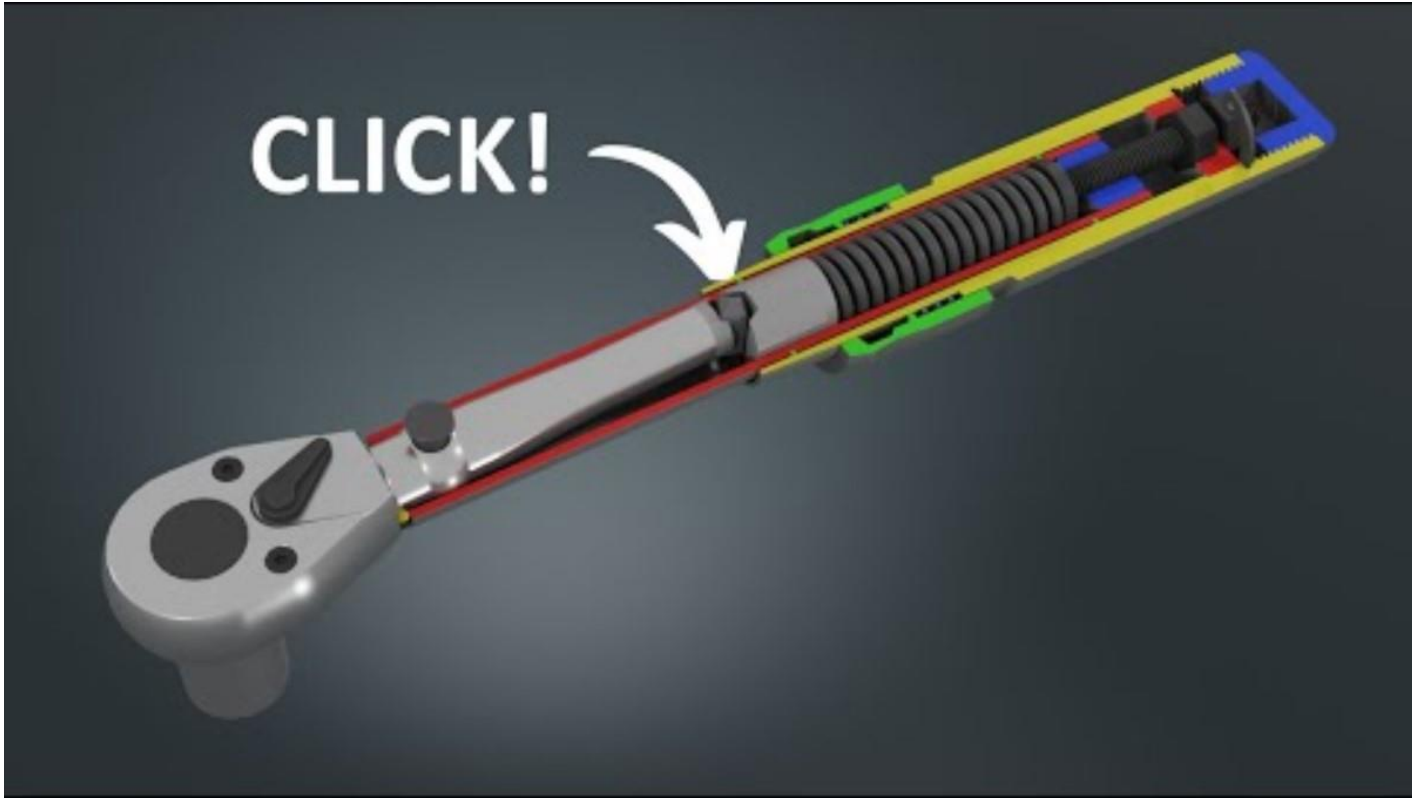
Tools

Tools

Torque wrench



Tools



Tools

T-Handle socket wrench

ComforPlanet
Cooling, Heating, Life

Tools

Ratchet spanner

Tools

Combination ratchet spanner

Tools

Double-ended socket ratchet wrench (spike type)

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Bolt

Allen head bolt (= Allen key head bolt, Socket head bolt, Hex socket head bolt)

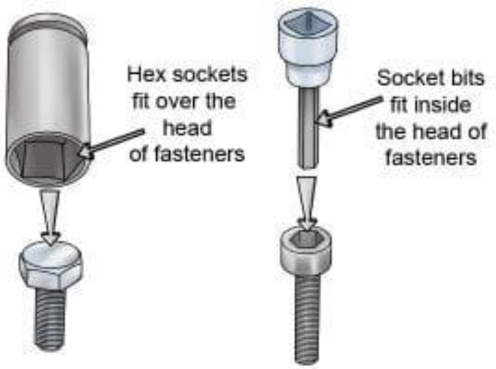
Tools

Allen wrench (= Allen key, Hex wrench, Hex key)

Ball-end hex wrenches

Tools

Allen socket (= Allen key socket, Hex key socket)



Tools

Impact wrench (= Air impact wrench,

Pneumatic impact wrench)

Tools

Pneumatic derusting brush

Tools

Bench grinder with grinding wheel and wire brush

Never use glove for rotating machine

Mind finger cutting accident

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Tools

Tools

Hammer

Steel hammer

Copper hammer

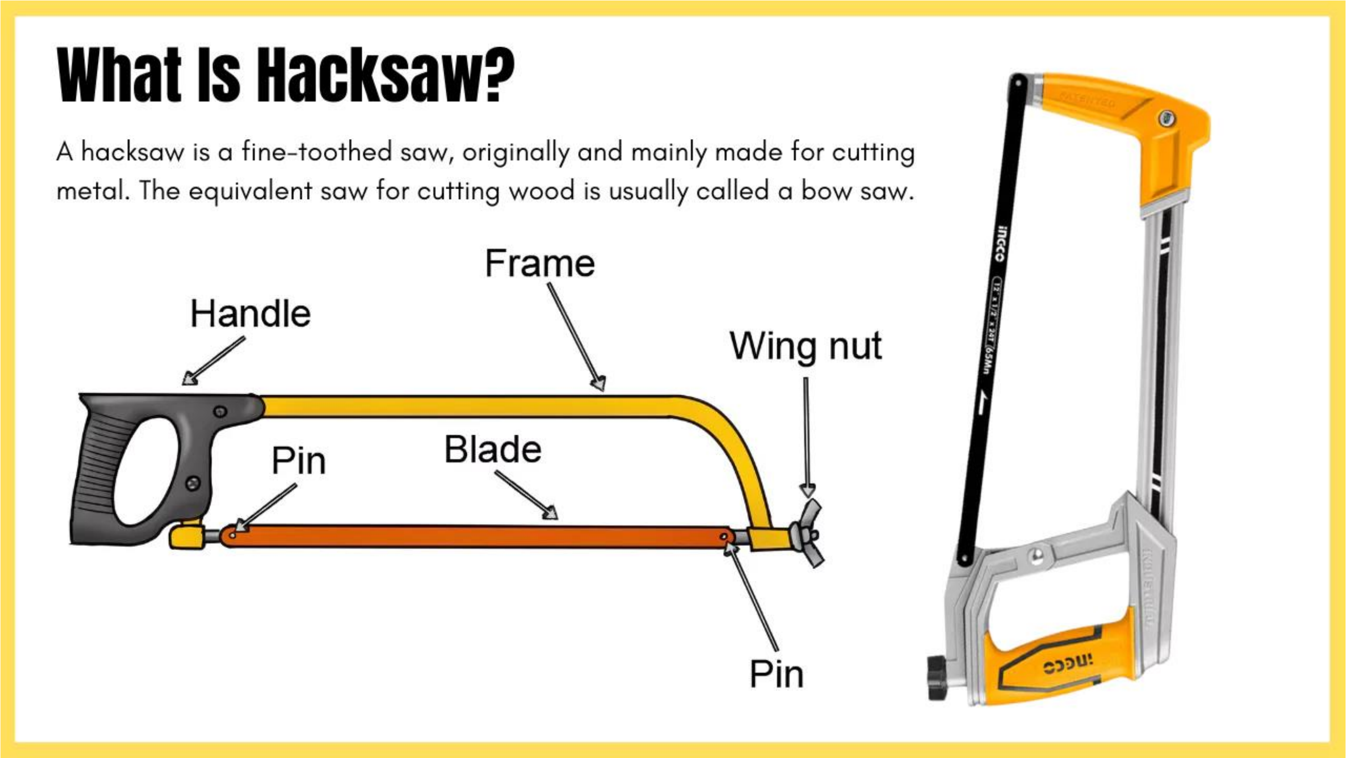
Rubber hammer

Teflon hammer

Plastic hammer

Tools

Hacksaw



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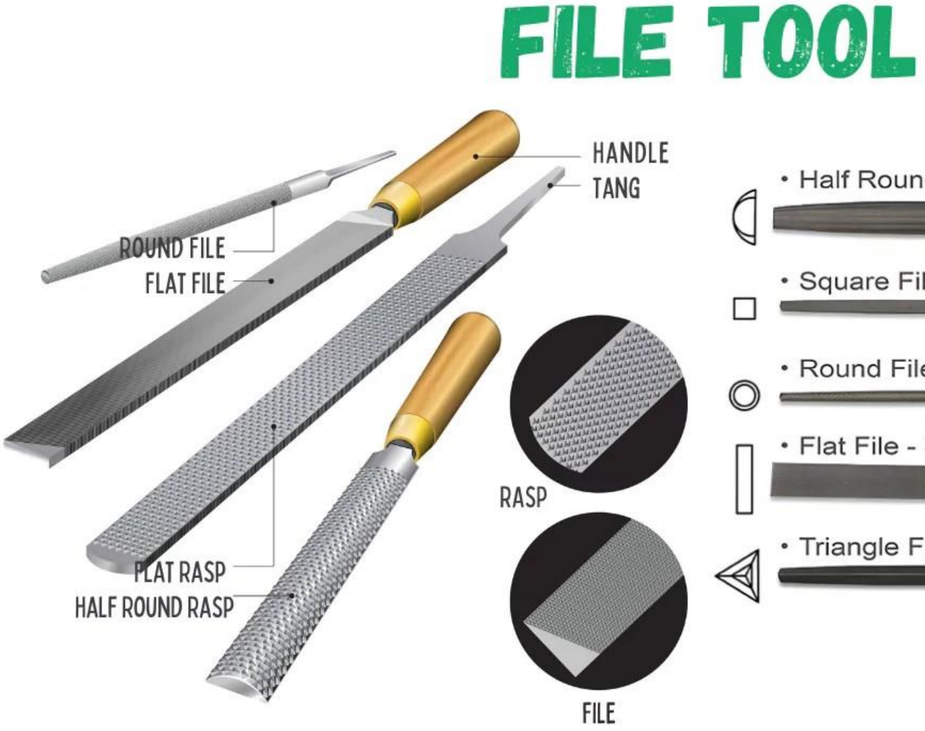
Tools

File



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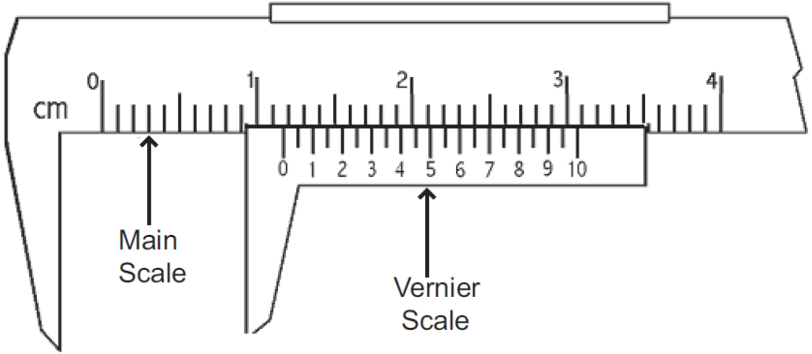
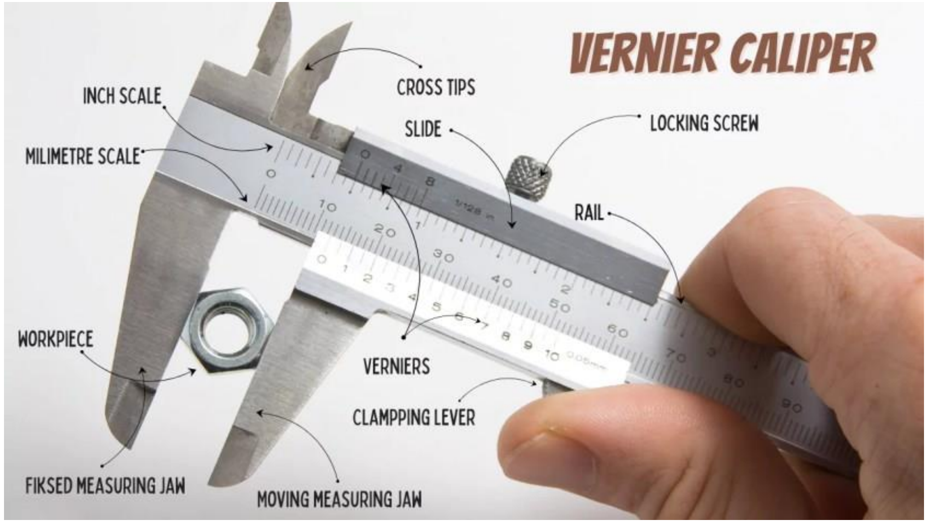
Tools

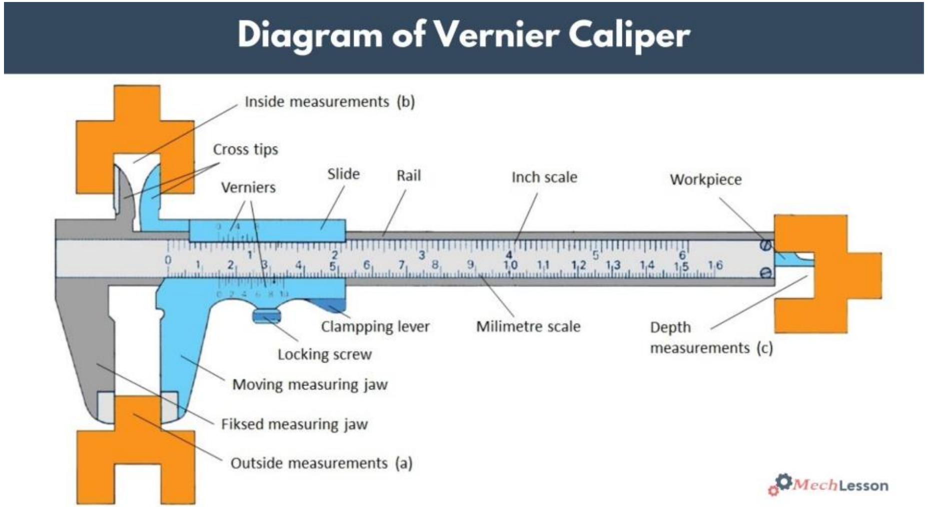
Sand paper

Tools

Vernier caliper







Tools

Vise plier (= Vise grip plier, Locking plier)

Tools

Long nose plier

Tools

Combination plier

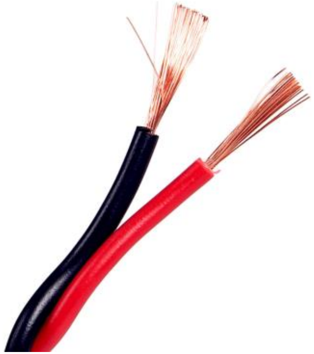
92501078

Tools

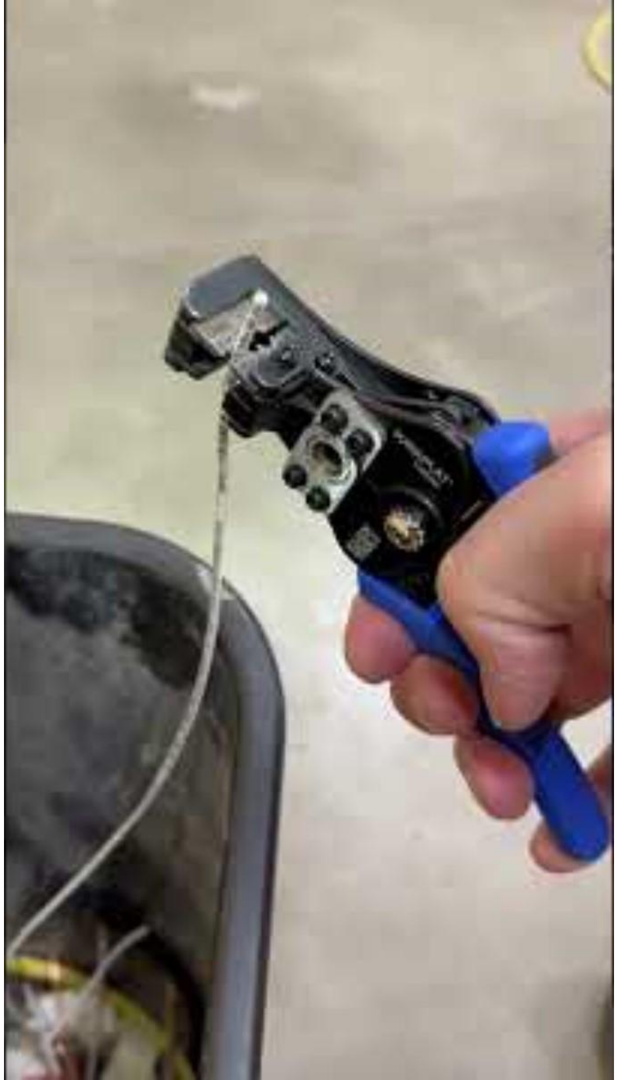
Diagonal cutting plier (= Nipper)

Tools

Electric wire stripper



Tools



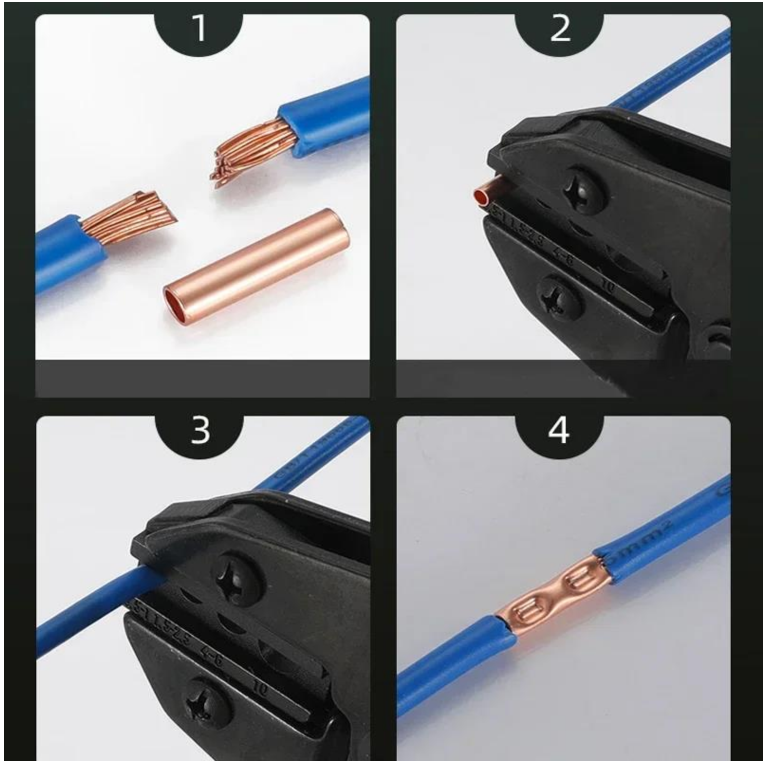
Tools

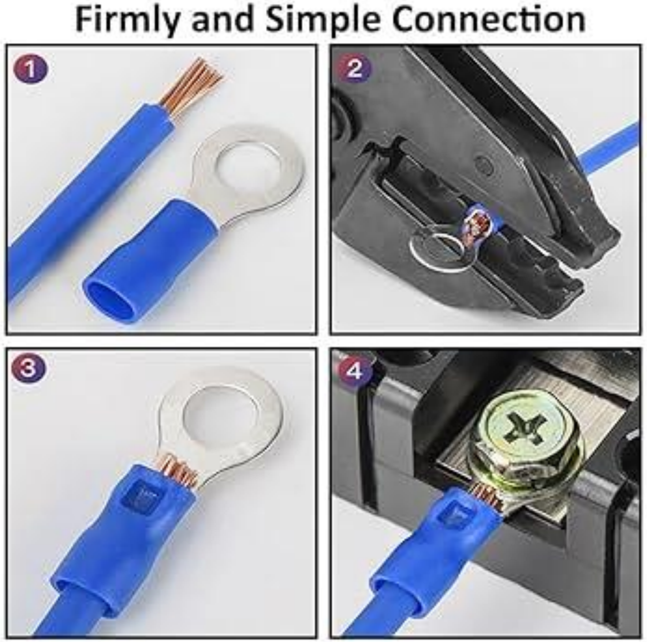
Ratchet crimping tool

(Crimping plier, Cable shoe plier)

Tools

Ratchet crimping tool (Crimping plier, Cable shoe plier)





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Tools

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Tools

Circlip plier

Tools

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Tools

Punching tool set

Tools

https://www.tiktok.com/@engineroominsight/video/7535494395844463879

Tools

Tools

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Tools

Hollow punch set

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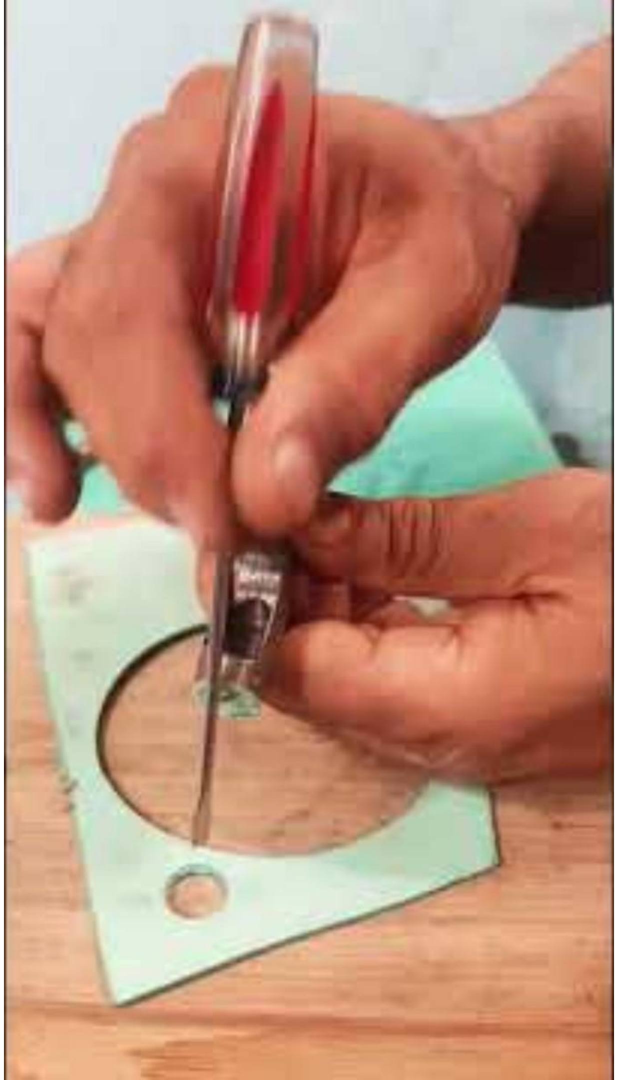
Tools



Tools

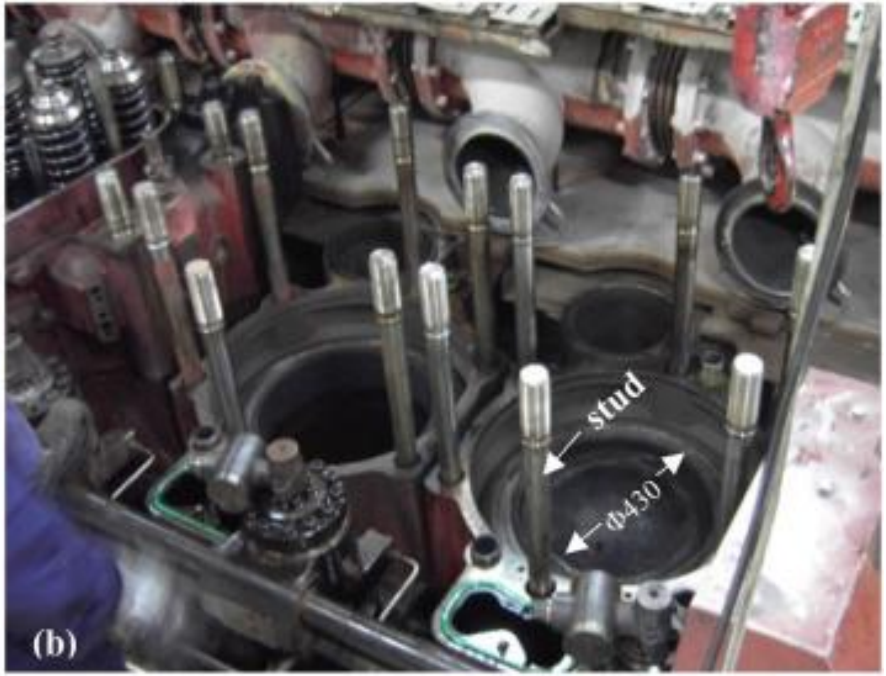
Digital multi tester

Tools

Digital insulation resistance tester (Digital ohm meter)

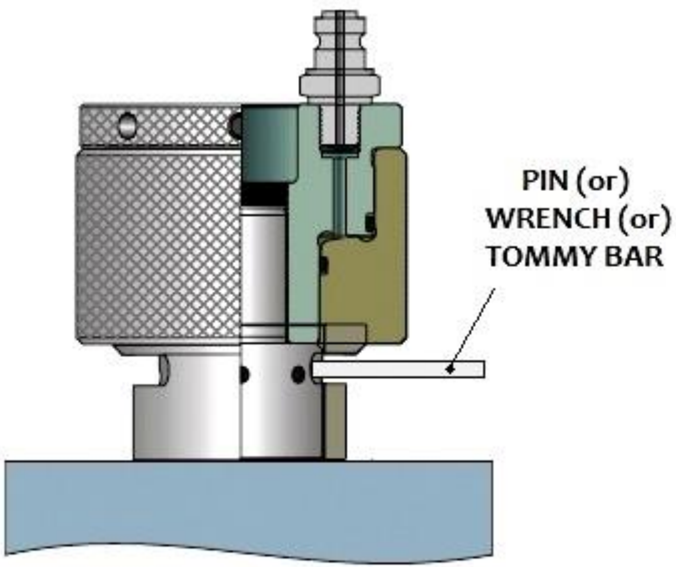
Tools

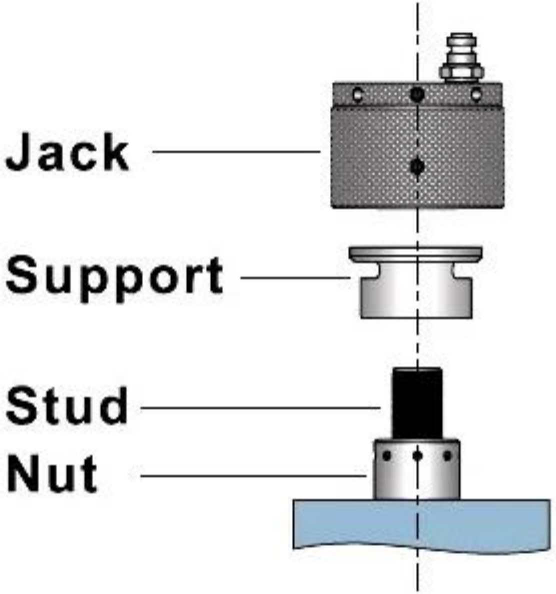
G/E cylinder head



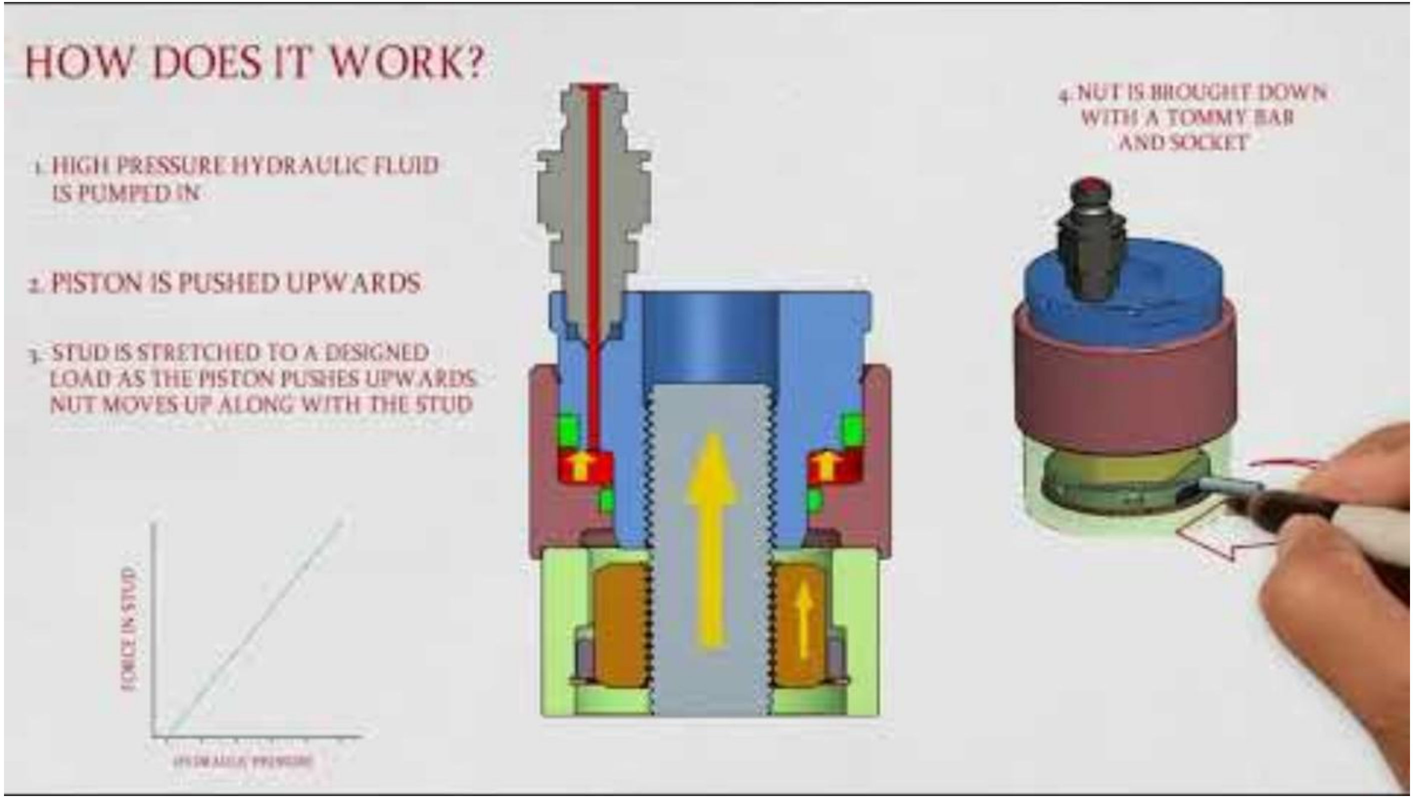
Tools

Hydraulic jack



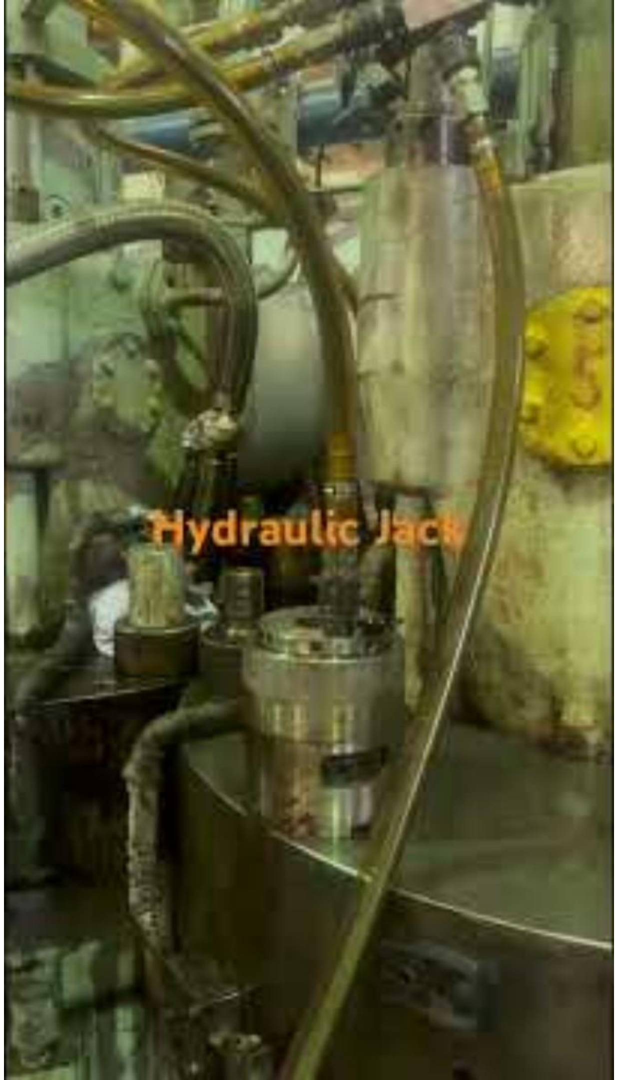


Tools


HOW DOES IT WORK?
1. HIGH PRESSURE HYDRAULIC FLUID IS PUMPED IN
2. PISTON IS PUSHED UPWARDS
3. STUD IS STRETCHED TO A DESIGNED LOAD AS THE PISTON PUSHES UPWARDS NUT MOVES UP ALONG WITH THE STUD
4. NUT IS BROUGHT DOWN WITH A TOMMY BAR AND SOCKET
FORCE IN STUD

Tools

Tools



Apprentice engineer (A/E) Duties

In reality, an apprentice engineer is neither responsible for nor in charge of anything done on a ship.

However, I would like to introduce the work that an apprentice engineer would typically do on a ship. This may vary slightly from company to company and vessel to vessel, but I personally don't think it will vary much.

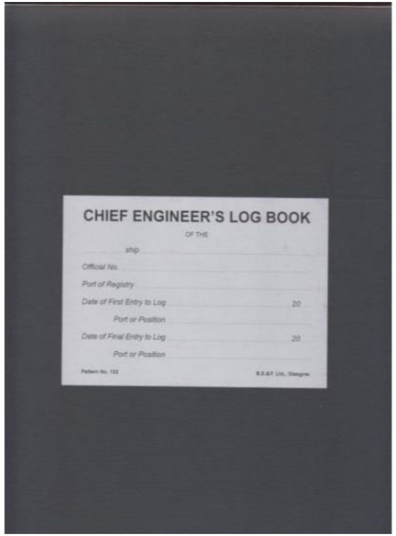
One of the most important duties of an apprentice engineer is writing a log book (= Chief engineer's log book).

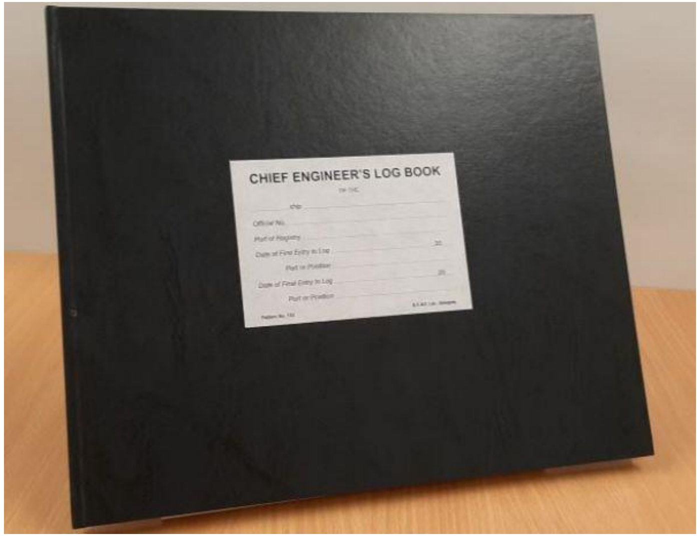
The log book written at 12:00 is usually written by the third-class engineer, and at 20:00, it is written by the on-duty engineer (sailing: $3^{rd} \rightarrow 2^{nd} \rightarrow 1^{st}$, in port: $3^{rd} \rightarrow 2^{nd}$).

However, sometimes the log book may be filled out by an apprentice engineer at 12:00 and 20:00.

Apprentice engineer (A/E) Duties

A log book looks like this, and while each ship and company has its own unique log book, its basic purpose remains the same.





Apprentice engineer (A/E) Duties

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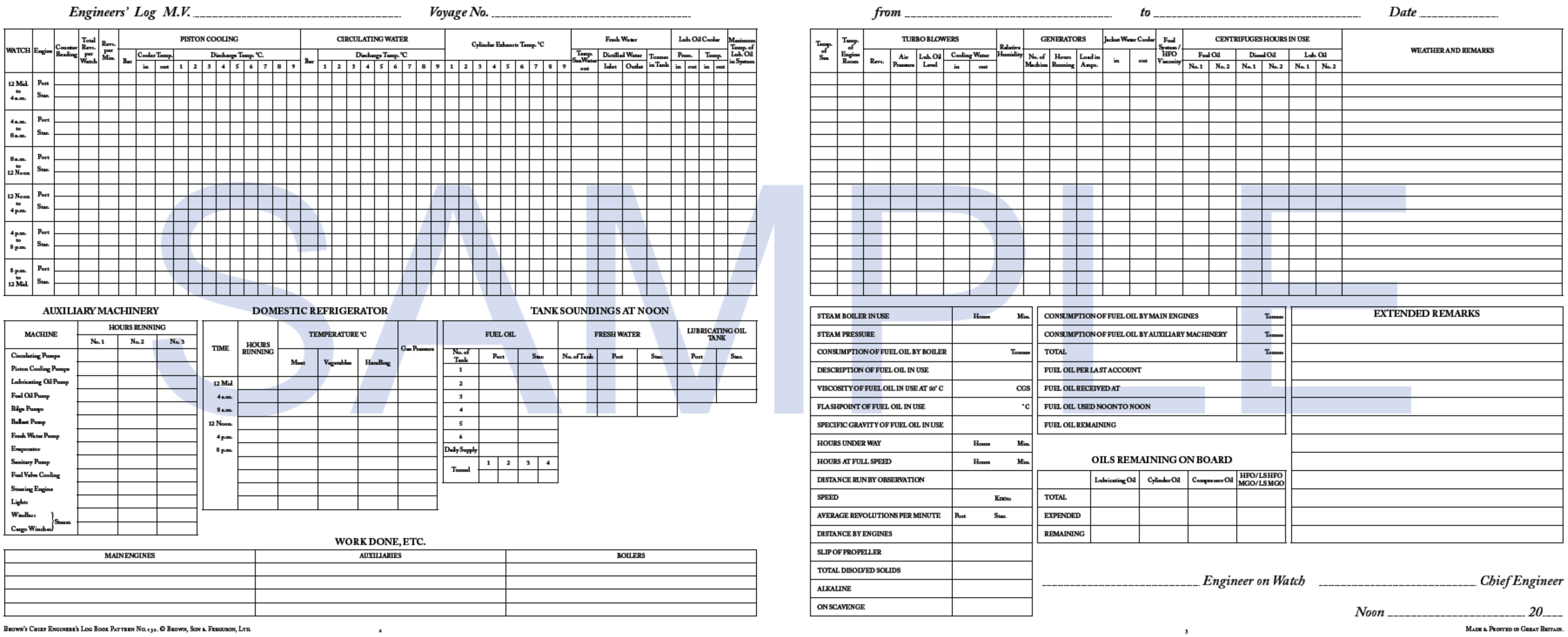

Engineers' Log M.V. ______ Voyage No. ______ from ______ to ______ Date ______

WATCH | Engine | Counter Reading | Total Revs. per Watch | Revs. per Min. | PISTON COOLING | CIRCULATING WATER | Cylinder Exhaust Temp. °C | Fresh Water | Lub. Oil Cooler | Maximum Temp. of Lub. Oil in System

12 Mid. to 4 a.m. | 4 a.m. to 8 a.m. | 8 a.m. to 12 Noon | 12 Noon to 4 p.m. | 4 p.m. to 8 p.m. | 8 p.m. to 12 Mid. — Port / Star.

Temp. of Sea | Temp. of Engine Room | TURBO BLOWERS | Relative Humidity | GENERATORS | Jacket Water Cooler | Fuel System / HFO Viscosity | CENTRIFUGES HOURS IN USE | WEATHER AND REMARKS

AUXILIARY MACHINERY — MACHINE | HOURS RUNNING No. 1, No. 2, No. 3: Circulating Pumps, Piston Cooling Pumps, Lubricating Oil Pump, Fuel Oil Pump, Bilge Pumps, Ballast Pump, Fresh Water Pump, Evaporator, Sanitary Pump, Fuel Valve Cooling, Steering Engine, Lights, Windlass, Cargo Winches (Steam)

DOMESTIC REFRIGERATOR — TIME | HOURS RUNNING | TEMPERATURE °C: Meat, Vegetables, Handling | Gas Pressure — 12 Mid, 4 a.m., 8 a.m., 12 Noon, 4 p.m., 8 p.m.

TANK SOUNDINGS AT NOON — FUEL OIL | FRESH WATER | LUBRICATING OIL TANK

WORK DONE, ETC. — MAIN ENGINES | AUXILIARIES | BOILERS

STEAM BOILER IN USE; STEAM PRESSURE; CONSUMPTION OF FUEL OIL BY BOILER; DESCRIPTION OF FUEL OIL IN USE; VISCOSITY OF FUEL OIL IN USE AT 50° C; FLASHPOINT OF FUEL OIL IN USE; SPECIFIC GRAVITY OF FUEL OIL IN USE; HOURS UNDER WAY; HOURS AT FULL SPEED; DISTANCE RUN BY OBSERVATION; SPEED; AVERAGE REVOLUTIONS PER MINUTE; DISTANCE BY ENGINES; SLIP OF PROPELLER; TOTAL DISOLVED SOLIDS; ALKALINE; ON SCAVENGE

CONSUMPTION OF FUEL OIL BY MAIN ENGINES; CONSUMPTION OF FUEL OIL BY AUXILIARY MACHINERY; TOTAL; FUEL OIL PER LAST ACCOUNT; FUEL OIL RECEIVED AT; FUEL OIL USED NOON TO NOON; FUEL OIL REMAINING

OILS REMAINING ON BOARD — Lubricating Oil | Cylinder Oil | Compressor Oil | HFO/LSHFO MGO/LSMGO — TOTAL, EXPENDED, REMAINING

EXTENDED REMARKS

______ Engineer on Watch ______ Chief Engineer

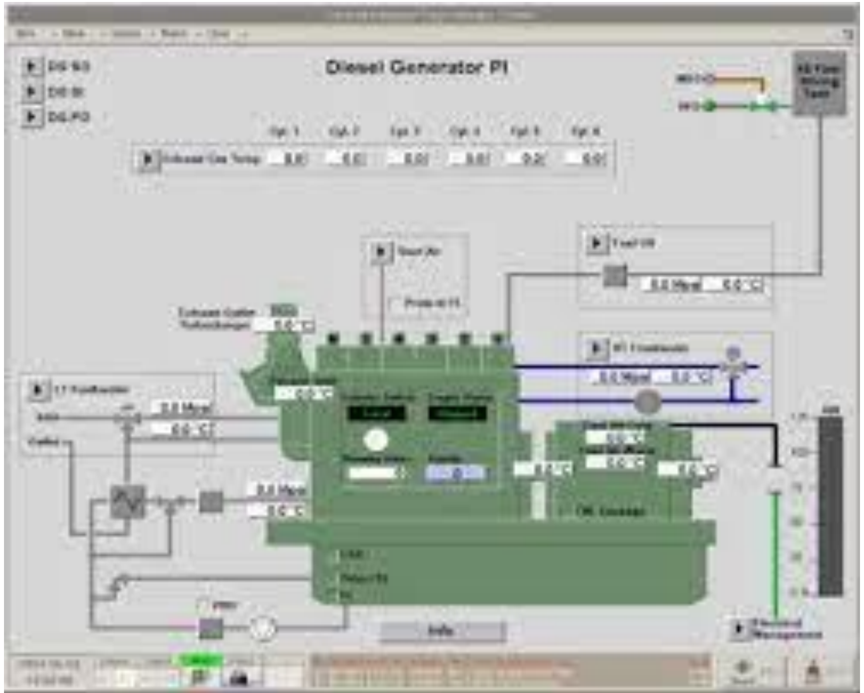
Noon ______ 20___

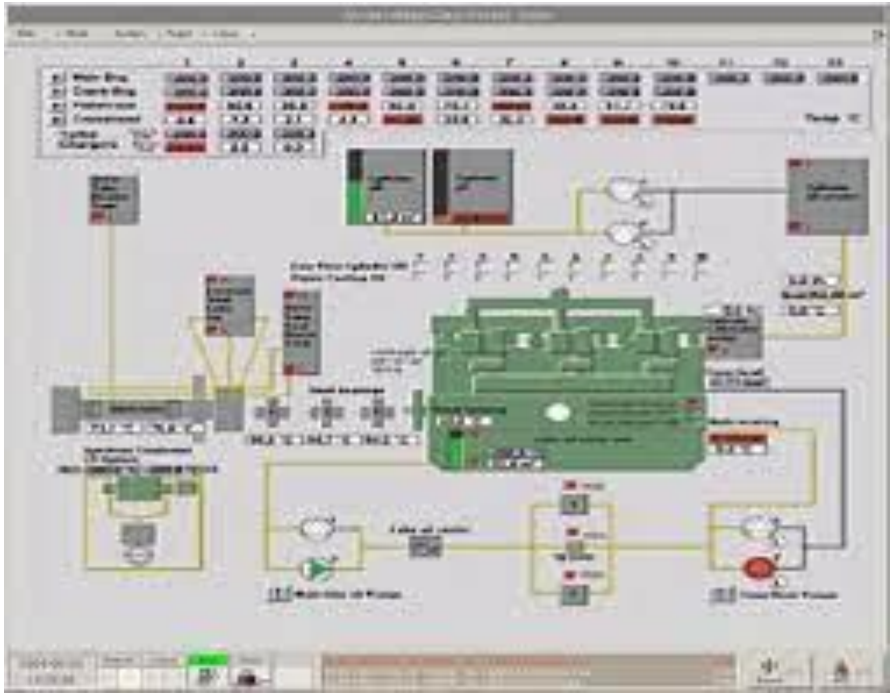
Brown's Chief Engineer's Log Book Pattern No. 131. © Brown, Son & Ferguson, Ltd. Made & Printed in Great Britain.

SAMPLE

Apprentice engineer (A/E) Duties

Alarm monitoring system (AMS) in the engine control room (ECR)





Apprentice engineer (A/E) Duties

Thermometers and pressure gauges attached to the machinery

Pressure gauges

Digital value based on sensor, not gauge value

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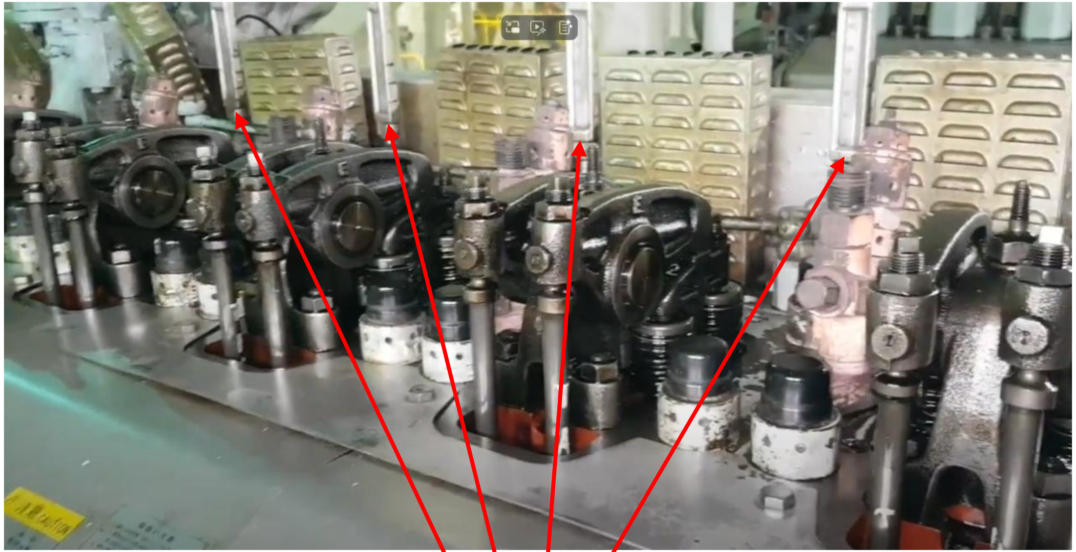
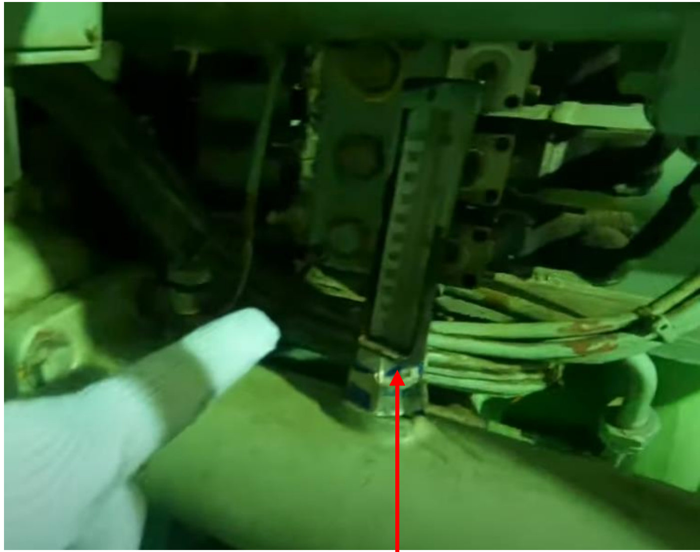
Apprentice engineer (A/E) Duties

Thermometers and pressure gauges attached to the machinery



Thermometers



Thermometers

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Apprentice engineer (A/E) Duties

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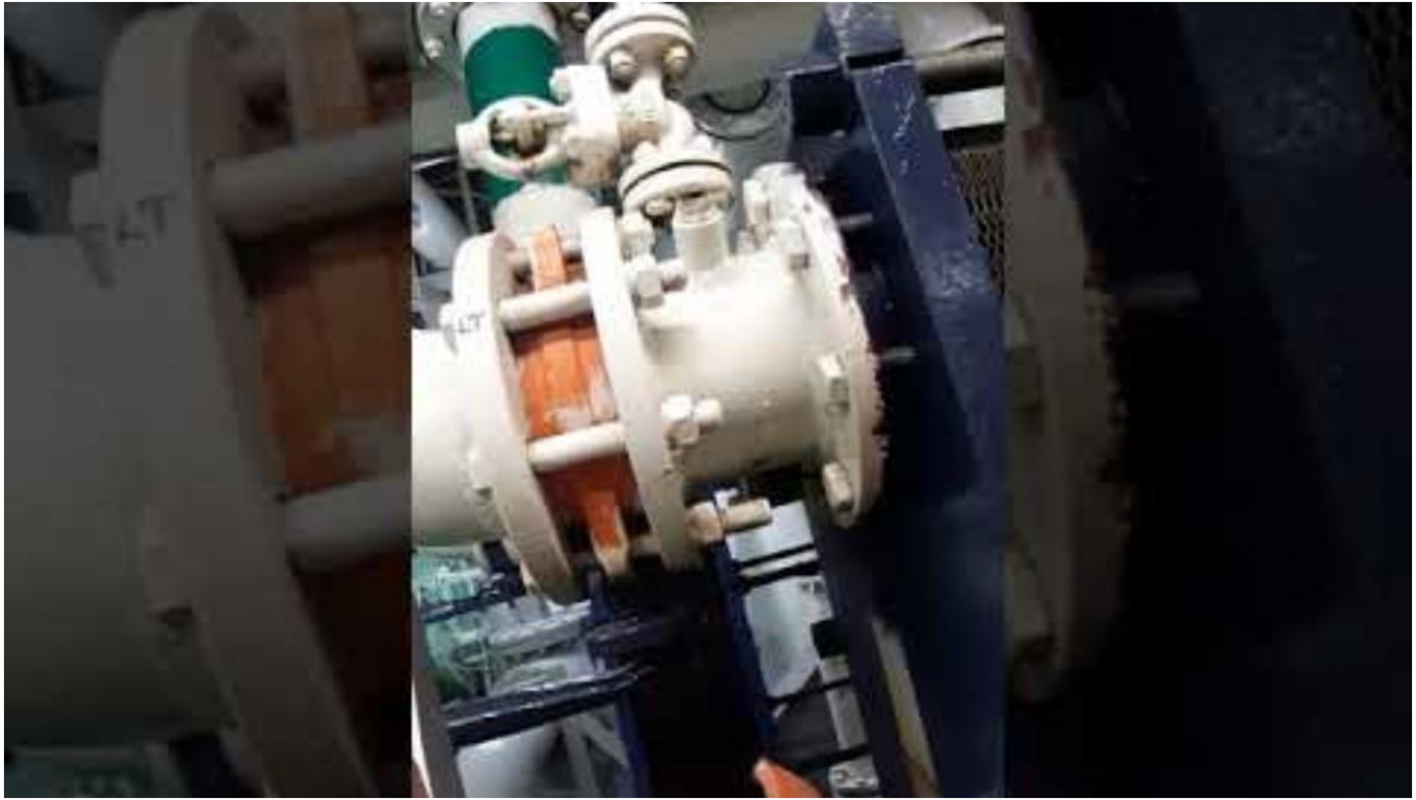
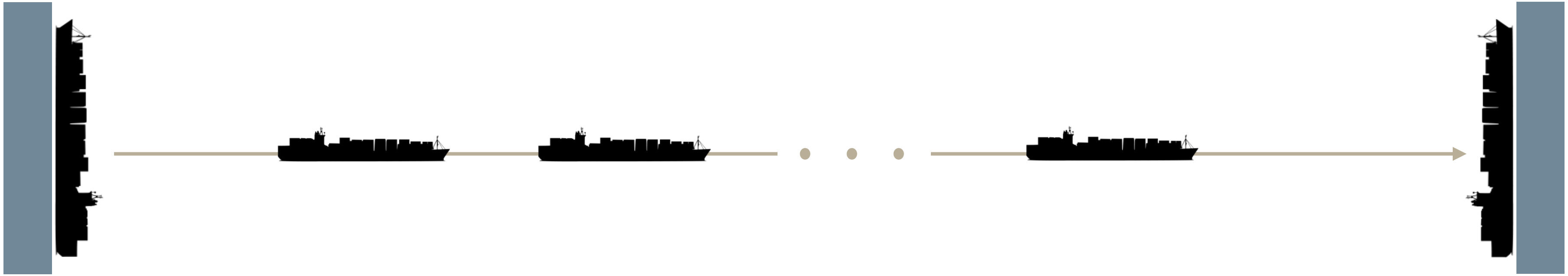
Apprentice engineer (A/E) Duties

Apprentice engineer (A/E) Duties



Apprentice engineer (A/E) Duties

Port to Port



Stand by (S/B)

Pilot onboard

Pilot offboard

Run-up (R/up)engine

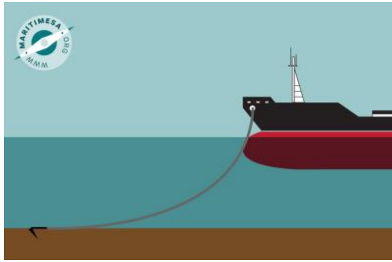
Stand by (S/B)

Pilot onboard

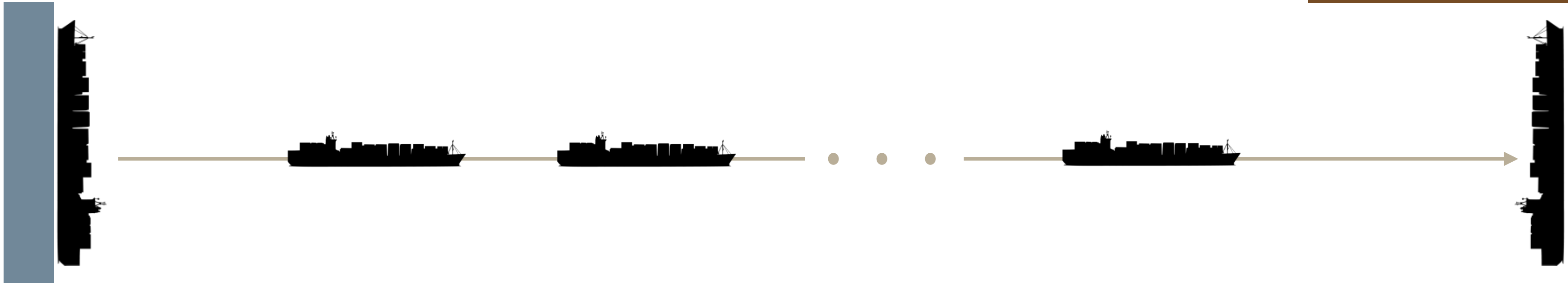
Finished with engine (F/W)

Pilot offboard

Apprentice engineer (A/E) Duties



Port to Anchorage



Stand by (S/B)

Pilot onboard

Pilot offboard

Run-up (R/up)engine

Stand by (S/B)

Finished with engine (F/W)

Apprentice engineer (A/E) Duties

Anchorage to Port

Stand by (S/B)

Pilot onboard

Finished with engine (F/W)

Pilot offboard

Apprentice engineer (A/E) Duties

In the case of very old ships, a duty system was implemented and the duty system was maintained as follows.

1st engineer: 04:00~08:00 / 16:00~20:00

2nd engineer: 00:00~04:00 / 12:00~16:00

3rd engineer: 08:00~12:00 / 20:00~24:00

However, as ships became automated, the watch system was stopped and the unattended machinery space (UMS) system was started. You can assume that all the ships you will be boarding are using the UMS system.

UMS is also called as unattended machinery area (UMA).

Apprentice engineer (A/E) Duties

The UMS system operates with 1st, 2nd, and 3rd engineers working together from 08:00 to 17:00, and then a duty system (3rd → 2nd → 1st) is implemented during the evening and early morning hours (17:00 to 08:00). (Times may vary slightly depending on the company and vessel.)

When you use the UMS system, you will be required to write a UMS checklist at 16:00 to check and record the status of most machines in the engine room.

The UMS checklist, like the log book, reads and records the values in the AMS and on-site (gauges).

Apprentice engineer (A/E) Duties

HANJIN SHIPPING

ENGINE DEPARTMENT U.M.A. CHECK LIST

(기관실 UMA 유지시 매일 1600시에 순찰로 필히 기재)

M/V HANJIN BREMERHAVEN VOY. NO: 0035W&E

No	CHECK POINT			DATE 05.22	DATE 05.23	DATE 05.24	DATE 05.18	DATE 05.19	DATE 05.20	DATE 05.21
1	AIR COND. FAN ROOM	A/C CENT. FAN AMP(N:34) / COND.		/	/	/	/	/	/	/
		AIR IN / OUT TEMP.		/	/	/	/	/	/	/
		R-404A OUT TEMP No.1/No.2		/	/	/	/	/	/	/
		LGSPCONDITION/DEGREE OF SUPERHEAT		/	/	/	/	/	/	/
2	REF. CHAMBER TEMP	MEAT ROOM(N:-20)								
		FISH ROOM(N:-20)								
		VEGETABLE ROOM(N:3)								
3	FUNNEL EXH. GAS CONDITION	M/E								
		G/E								
4	E/R FAN ROOM CONDITION (PORT / STBD SIDE)			/	/	/	/	/	/	/
5	BOILER & INCI. EXH. PIPE DRAIN TANK CONDITION									
6	INCINERATOR D.O TK / W.O TANK LEVEL			/	/	/	/	/	/	/
7	EXH.GAS ECONOMIZER MANOMETER									
8	INCINERATOR CONTROL PANEL CONDITION									
9	AUX. BOILER DRUM WATER LEVEL(NL)									
10	AUX. BOILER	DRUM STEAM PRESS.(5.0-7.0)								
		F.O TEMP(N:115) / PANEL COND.		/	/	/	/	/	/	/
		CHEMI. DOSING TANK LEVEL(1/5↑)								
11	H.T C.F.W. EXPANSION TK LEVEL (1.2↑)									
12	L.T C.F.W. EXPANSION TK LEVEL (0.8↑)									
13	WORK SHOP & STORE ROOM ARRANGEMENT									
14	WORK SHOP UNIT COOLER	COND. / TEMP		/	/	/	/	/	/	/
		PRESS. SUC. / DISCH.(N:0.6/1.8)		/	/	/	/	/	/	/
15	CYL OIL MEASURING TANK LEVEL									
16	PROV. REF. COMP. UNIT	IN RUN / MOTOR AMPARE		/	/	/	/	/	/	/
		LEVEL	COMP. L.O(N:1/2)							
			CONDENSER R-404A(N:1/3)							
		PRESS	SUC./DISCH.(N:1.0/16.5)	/	/	/	/	/	/	/
17	No.1 AIR COND. COMP UNIT	LEVEL	COMP. L.O(N:1/2)							
			CONDENSER R-404A(N:1/3)							
		PRESS	SUC./DIS(N:5.0/18.0)	/	/	/	/	/	/	/
			LO. (N:4.0)							
			MOTOR AMPARE							
18	No.2 AIR COND. COMP UNIT	LEVEL	COMP. L.O(N:1/2)							
			CONDENSER R-404A(N:1/3)							
		PRESS	SUC./DIS(N:5.0/18.0)	/	/	/	/	/	/	/
			LO. (N:4.0)							
			MOTOR AMPARE							
19	M/E EXH. PIPE DRAIN TANK CONDITION									
20	MINERALIZER / STERILIZER CONDITION			/	/	/	/	/	/	/
21	HOT W. CIRC. PUMP CONDITION / DIS. PRESS.			/	/	/	/	/	/	/
22	CALORIFIER CONDITION / TEMP(70↓)			/	/	/	/	/	/	/
	작성자 직책 & SIGN			/	/	/	/	/	/	/
	CHIEF ENGINEER'S SIGNATURE									

No	CHECK POINT				DATE 05.15	DATE 05.16	DATE 05.17	DATE 05.18	DATE 05.19	DATE 05.20	DATE 05.21
23	MAIN ENGINE	EXH. GAS / J.C.F.W TEMP 400↓ / 90 ± 3		NO.1	/	/	/	/	/	/	/
				NO.2	/	/	/	/	/	/	/
				NO.3	/	/	/	/	/	/	/
				NO.4	/	/	/	/	/	/	/
				NO.5	/	/	/	/	/	/	/
				NO.6	/	/	/	/	/	/	/
				NO.7	/	/	/	/	/	/	/
				NO.8	/	/	/	/	/	/	/
				NO.9	/	/	/	/	/	/	/
				NO.10	/	/	/	/	/	/	/
				NO.11	/	/	/	/	/	/	/
				NO.12	/	/	/	/	/	/	/
		CYL. COVER F.O, L.O, AIR LEAKAGE									
		RAIL UNIT LEAKAGE CONDITION (FWD/AFT)			/	/	/	/	/	/	/
		M/E TOP BRACING PRESS. (N:80)(FWD/AFT)		PORT	/	/	/	/	/	/	/
				STBD	/	/	/	/	/	/	/
		SPRING AIR DRAIN TRAP (FWD/AFT)			/	/	/	/	/	/	/
		NO.1 T/C	AIR FILTER MANOMETER(70↓)								
			EXH.GAS IN/OUT T.(N:430/340)		/	/	/	/	/	/	/
			L.O TEMP. / PRESS.(75↓/0.9↑)		/	/	/	/	/	/	/
		NO.2 T/C	AIR FILTER MANOMETER(70↓)								
			EXH.GAS IN/OUT T.(N:430/340)		/	/	/	/	/	/	/
			L.O TEMP. / PRESS.(75↓/0.9↑)		/	/	/	/	/	/	/
		NO.3 T/C	AIR FILTER MANOMETER(70↓)								
			EXH.GAS IN/OUT T.(N:430/340)		/	/	/	/	/	/	/
			L.O TEMP. / PRESS.(75↓/0.9↑)		/	/	/	/	/	/	/
		C.O LUB. WORKING CONDITION									
		SERVO OIL AUTO FILTER COUNT/△P COND.			/	/	/	/	/	/	/
		F.O IN / OUT PRESSURE			/	/	/	/	/	/	/
		F.O INLET TEMP.									
		SUPPLY UNIT	FUEL PUMP LEAK / COND		/	/	/	/	/	/	/
			SERVO OIL PUMP LEAK/COND		/	/	/	/	/	/	/
		CONTROL OIL PUMP LEAK / COND			/	/	/	/	/	/	/
		No.1 AIR COOLER	AIR DIFF. PRESS (200↓)								
			TEMP. AIR IN/OUT(N:150/45)		/	/	/	/	/	/	/
			TEMP. C.W IN/OUT(75↓)		/	/	/	/	/	/	/
		No.2 AIR COOLER	AIR DIFF. PRESS (200↓)								
			TEMP. AIR IN/OUT(N:150/45)								
			TEMP. C.W IN/OUT(75↓)								
		No.3 AIR COOLER	AIR DIFF. PRESS (200↓)								
			TEMP. AIR IN/OUT(N:150/45)		/	/	/	/	/	/	/
			TEMP. C.W IN/OUT(75↓)		/	/	/	/	/	/	/
		SPRING AIR(CONT.AIR/ST-BY AIR) PRESS(N:6.8)			/	/	/	/	/	/	/
24	WATER MIST FIRE PUMP UNIT CONDITION										
25	BOILER FEED WATER SALINITY										
26	INSPECTION CHAMBER CONDITION										
27	CASCADE TANK LEVEL / TEMP.(2.0↑/N:40`C)				/	/	/	/	/	/	/
28	DRAIN COOLER	TEMP. DRAIN IN / OUT			/	/	/	/	/	/	/
		TEMP. C.W IN / OUT			/	/	/	/	/	/	/
29	VACUUMARATOR START/STOP PRESS. / TIME				/	/	/	/	/	/	/

No	CHECK POINT		DATE 05.15	DATE 05.16	DATE 05.17	DATE 05.18	DATE 05.19	DATE 05.20	DATE 05.21
30	BOILER FEED WATER TANK LEVEL								
31	MAIN L.O COOLER	No.1 L.O TEMP IN / OUT(N:45)	/	/	/	/	/	/	/
		No.2 L.O TEMP IN / OUT(N:45)	/	/	/	/	/	/	/
32	No.1 CENTRAL F.W COOLER	S.W TEMP. IN / OUT	/	/	/	/	/	/	/
		F.W TEMP. IN / OUT(N:36)	/	/	/	/	/	/	/
33	No.2 CENTRAL F.W COOLER	S.W TEMP. IN / OUT	/	/	/	/	/	/	/
		F.W TEMP. IN / OUT(N:36)	/	/	/	/	/	/	/
34	CYL. OIL STORAGE TANK LEVEL								
35	F.W GEN.	SHELL VACUUM / TEMP(720/60)	/	/	/	/	/	/	/
		SALINITY ALARM SET/ACTUAL(10/)	10 /	10 /	10 /	10 /	10 /	10 /	10 /
		DISTILLATE PUMP PRESS.							
36	M/E J.C.F.W COOLER JACKET F.W IN / OUT								
37	M/E J.C.F.W PUMP	IN RUN							
		PRESS. IN / OUT (N:1.3/4.3)	/	/	/	/	/	/	/
38	CENTRAL CLG F.W PUMP	IN USE							
		PRESS. IN / OUT (N:1.1/3.8)	/	/	/	/	/	/	/
		PRESS. IN / OUT (N:1.1/3.8)	/	/	/	/	/	/	/
39	M.D.O SETT. & SERV. TANK DRAINING		/	/	/	/	/	/	/
40	CONTROL OIL PUMP IN RUN / AMP(N:28)		/	/	/	/	/	/	/
41	BILGE FIRE & G/S PUMP CONDITION								
42	BALLAST PUMP CONDITION								
43	MAIN CLG S.W PUMP	IN USE							
		PRESS. IN / OUT (N:0.5/2.6)	/	/	/	/	/	/	/
		PRESS. IN / OUT (N:0.5/2.6)	/	/	/	/	/	/	/
44	F.W GENERATOR EJECTOR PUMP PRESS.OUT(N:4.2)/AMP		/	/	/	/	/	/	/
45	M.G.P.S	CU AMP No.1 / No.2 / No3							
		AL AMP No.1 / No.2							
		IN USE / FLOW RATE (15↑)	/	/	/	/	/	/	/
46	AXIAL DETUNER	DRIVING END PRESS (4.0-5.0)	/	/	/	/	/	/	/
		FREE END PRESS (4.0-5.0)	/	/	/	/	/	/	/
		OIL PRESS. FOR DAMPER (4.0-5.0)	/	/	/	/	/	/	/
47	AIR COOLER DRAIN	No.3 AIR CLR DRAIN, WATER / OIL	/	/	/	/	/	/	/
		No.2 AIR CLR DRAIN, WATER / OIL	/	/	/	/	/	/	/
		No.1 AIR CLR DRAIN, WATER / OIL	/	/	/	/	/	/	/
48	AIR COOLER WATER DRAIN AIR VENT COND.								
49	POTABLE W. HYD. UNIT COND / PRESSURE(N:6.0)		/	/	/	/	/	/	/
50	F.W. HYD. UNIT COND / PRESSURE(N:6.0)		/	/	/	/	/	/	/
51	BLR FEED. WATER PUMP	IN RUN							
		PRESS. IN / OUT(N:0.5/14.5)	/	/	/	/	/	/	/
52	M/E L.O SUMP TANK LEVEL								
53	SHAFT GROUNDING SYSTEM (mV) (50↓)		/	/	/	/	/	/	/
54	SLUDGE PUMP & SLUDGE TRANS PUMP CONDITION								
55	M/E THRUST BEARING L.O TEMP(65↓)		/	/	/	/	/	/	/
56	M/E RESIDUE OIL TRAP CONDITION								
57	OILY BILGE PUMP CONDITION								
58	OILY WATER SEPARATOR CONDITION								
59	E/R BILGE WELL CONDITION/TRANSFER		/	/	/	/	/	/	/
60	E/R OTHER TANK CONDITION/TRANSFER		/	/	/	/	/	/	/

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Apprentice engineer (A/E) Duties

Calculation of fuel oil consumption for M/E, G/E, and BLR

A metric ton, also known as a tonne, is a unit of mass equal to 1,000 kilograms.

It is the standard unit of mass in the International System of Units (SI) and is widely used in international trade, shipping, and many countries around the world.

Apprentice engineer (A/E) Duties

Calculation of fuel oil consumption for M/E, G/E, and BLR

HOW TO CALCULATE THE CONSUMPTION OF FUEL OIL?															
GIVEN		SPECIFIC GRAVITY OF FUEL AT 15 *C								0.9850				COEF FACTOR	0.00067
		F.O SERVICE TANK TEMPERATURE *C								90					0.00065
		FLOWMETER READING 1 DEC 2019 @ 1200H LTRS								1830320				T1	15
		FLOWMETER READING 2 DEC 2019 @ 1200H LTRS								1849650					
		FM PRES	-	FM PREV		1849650	-	1830320		19330					
CONS IN M3	=				=				=		=	**19.33**			
		1000						1000		1000					
CORR. S.G	=	SG @ 15	-	T2	-	T1	X	COEF FACTOR							
	=	0.9850		90	-	15	X	0.00067							
	=	0.9850	-	75	X	0.00067						1M3	=	1000 LTRS	
	=	0.9850	-	0.05											
CORR. S.G	=	**0.9348**													
CONS IN MT	=	CORR. S.G	X	CONS IN M3											
	=	0.9348	X	19.33											
CONS IN MT	=	**18.07**													

You don’t need to calculate S.G.

2nd engineer will let you know.

This value continuously change after getting new fuel oil.

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Apprentice engineer (A/E) Duties

1,000kg = 1 ton

Calculation of fuel oil consumption for M/E, G/E, and BLR

Density of water ($\rho water$ = 1,000 kg/m^3 at 15℃)

Specific gravity (SG) measures a substance's density relative to the density of a reference substance, usually water. (SG has no units.)

Example: SG = 0.9850 → Density ρ = 0.9850 × 1,000 = 985 kg/m^3 → 985 / 1000 = 0.985 t/m^3

Therefore, fuel oil consumption formula is as follows.

Example: 0.985 x 1,000 ÷ 1,000 x (fuel oil consumption in m^3) = fuel oil consumption in t (ton)

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Apprentice engineer (A/E) Duties

Air reservoir tank drain

No. 1 air reservoir tank

No. 2 air reservoir tank

Water drain valves

Clockwise: Valve close
Counterclockwise: Valve open

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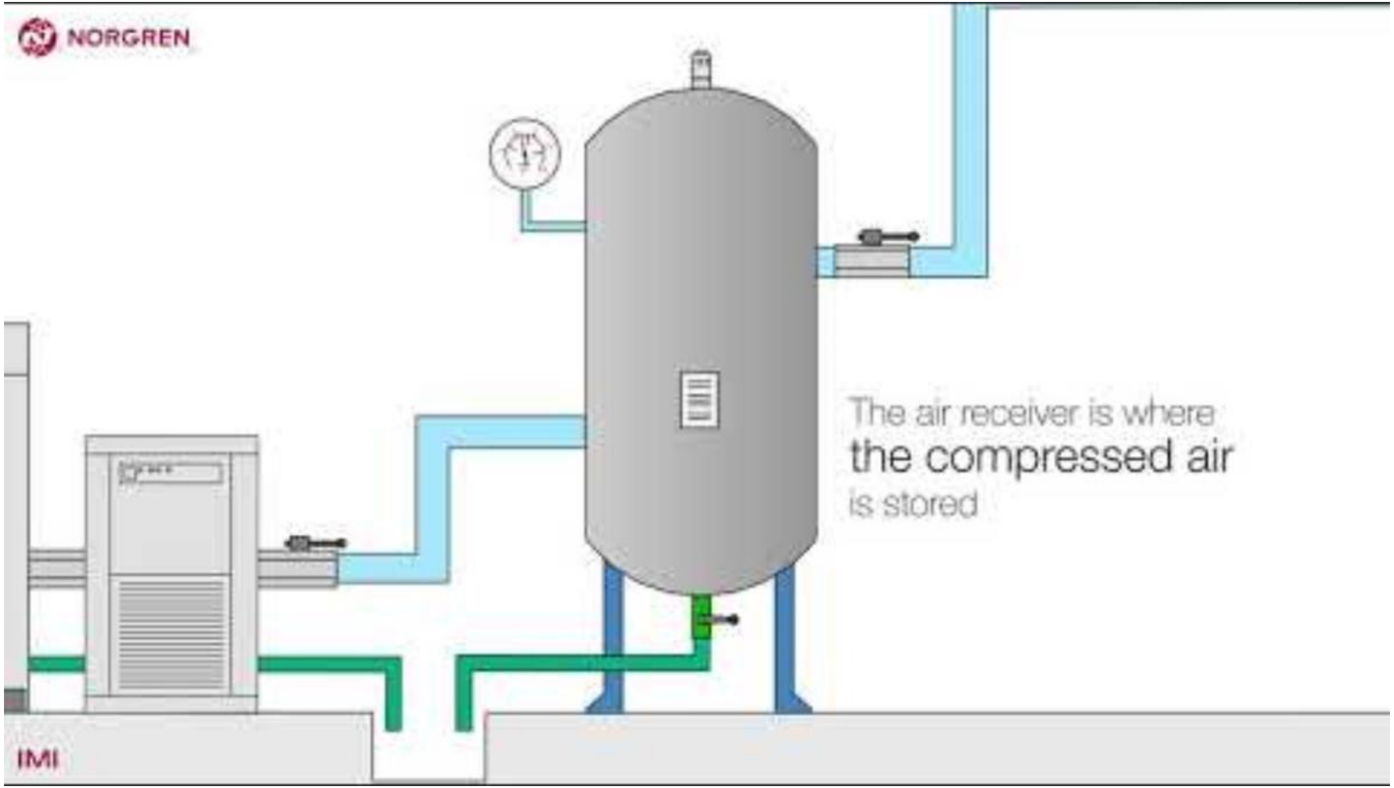
Apprentice engineer (A/E) Duties



Apprentice engineer (A/E) Duties

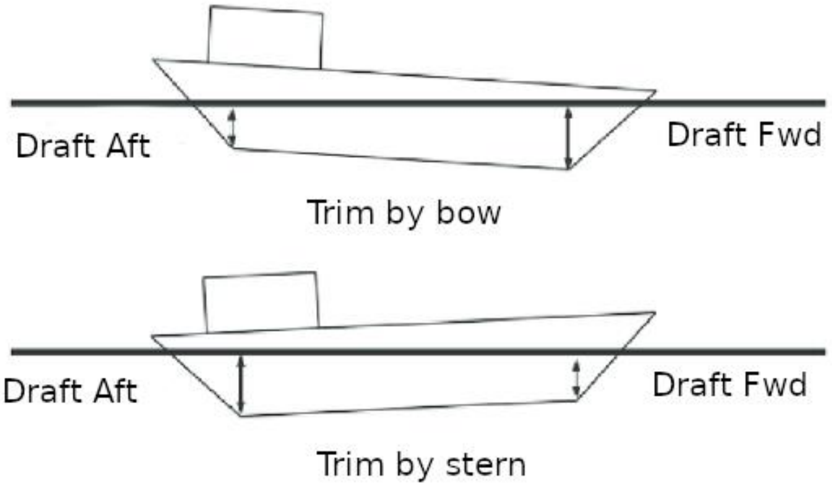
Fuel oil tank drain (F.O. tanks, HFO sett tank, HFO serv tank)

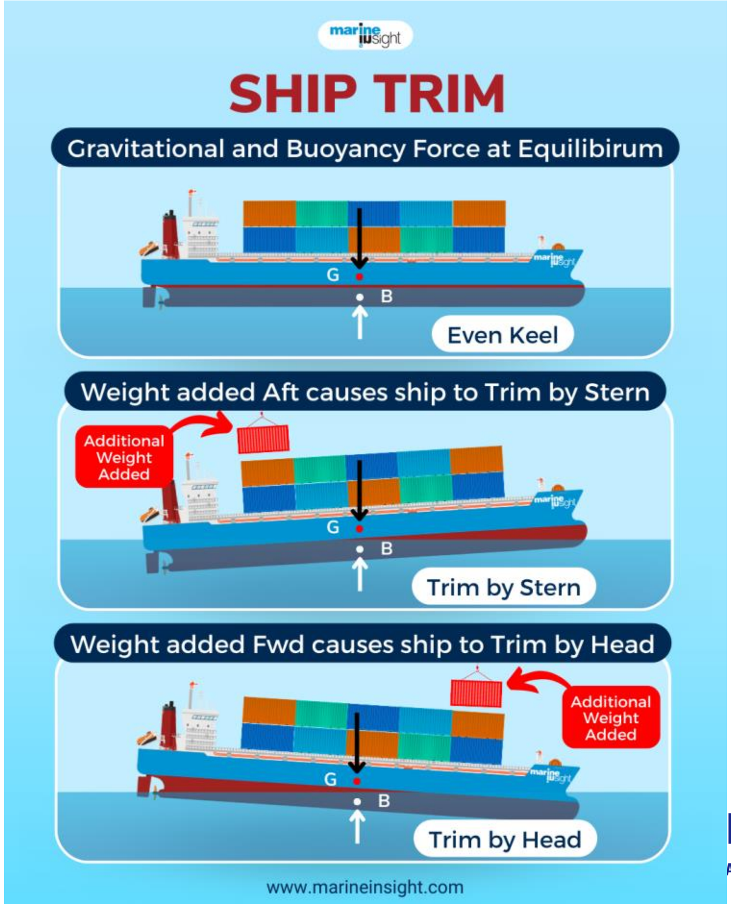
Apprentice engineer (A/E) Duties

Filling of fresh water to cooling water expansion tank

Apprentice engineer (A/E) Duties

Tank sounding (Fuel oil, Sludge oil, Bilge water, e.t.c.)





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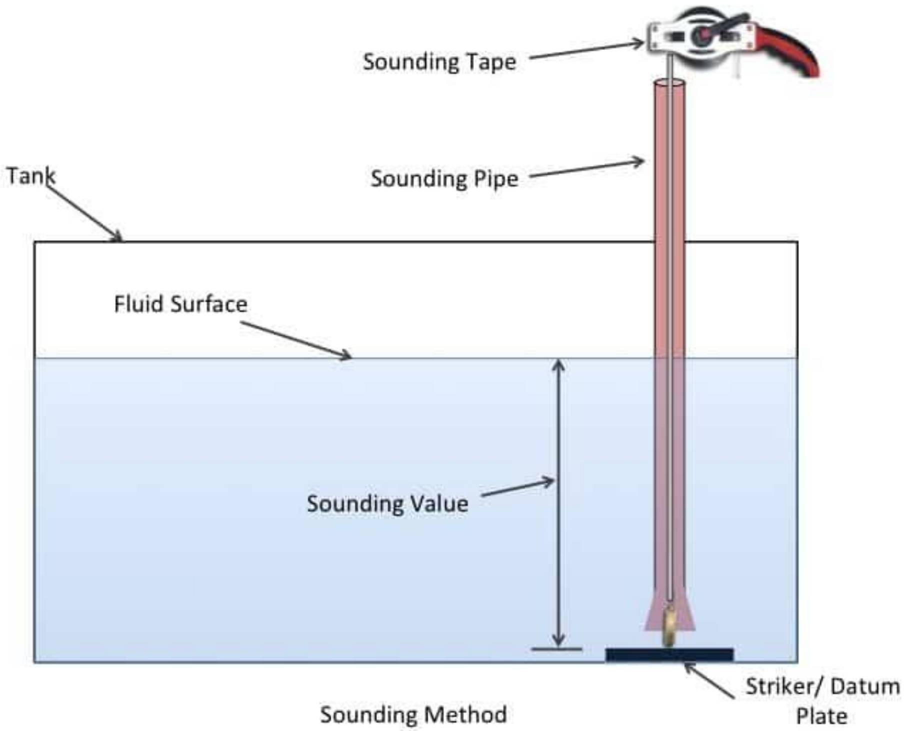
Apprentice engineer (A/E) Duties

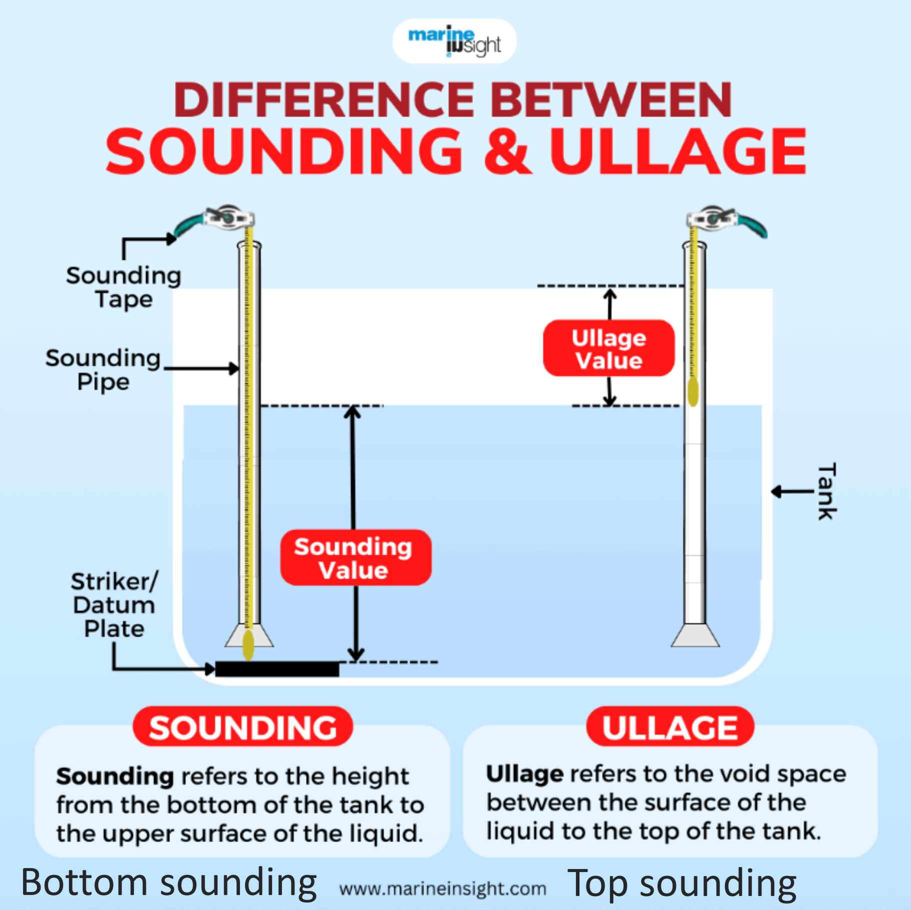
Tank sounding (Fuel oil, Sludge oil, Bilge water, e.t.c.)



Sounding Method

Apprentice engineer (A/E) Duties

Tank sounding (Fuel oil, Sludge oil, Bilge water, e.t.c.)

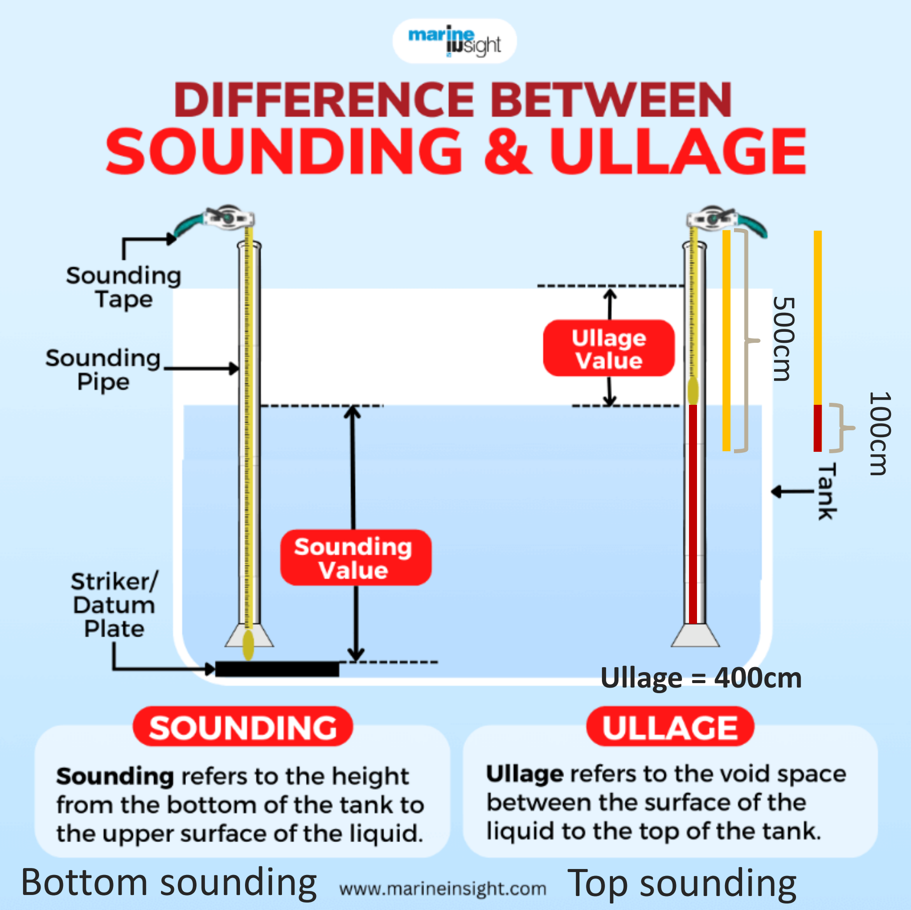


Bottom sounding

Top sounding

Apprentice engineer (A/E) Duties

Tank sounding (Fuel oil, Sludge oil, Bilge water, e.t.c.)



Apprentice engineer (A/E) Duties

Tank sounding (Fuel oil, Sludge oil, Bilge water, e.t.c.)

Quiz 1)

Sounding depth = 856 cm / Trim = By the stern -3.0 m

How much is the volume of tank => 537.34 m^3

Quiz 2)

Ullage depth = 463 cm / Trim = Even keel

How much is the volume of tank => 521.02 m^3

Always sounding depth + ullage depth is same.

*** TANK SOUNDING TABLES ***

SOUNDING DEPTH CM	TRIM HEAD 1.0M	EVEN KEEL 0.0M	CAPACITIES IN CUBIC METERS TRIM BY THE STERN -1.0M	-2.0M	-3.0M	-4.0M	-5.0M	ULLAGE DEPTH CM
800.	519.12	514.41	509.70	504.99	500.28	495.57	490.86	473.
802.	520.44	515.73	511.02	506.31	501.60	496.89	492.18	471.
804.	521.76	517.05	512.34	507.63	502.92	498.21	493.50	469.
806.	523.09	518.38	513.67	508.96	504.25	499.54	494.83	467.
808.	524.41	519.70	514.99	510.28	505.57	500.86	496.15	465.
810.	525.73	521.02	516.31	511.60	506.89	502.19	497.48	463.
812.	527.06	522.35	517.64	512.93	508.22	503.51	498.80	461.
814.	528.38	523.67	518.96	514.25	509.54	504.83	500.12	459.
816.	529.71	525.00	520.29	515.58	510.87	506.16	501.45	457.
818.	531.03	526.32	521.61	516.90	512.19	507.48	502.77	455.
820.	532.35	527.64	522.93	518.22	513.51	508.80	504.09	453.
822.	533.68	528.97	524.26	519.55	514.84	510.13	505.42	451.
824.	535.00	530.29	525.58	520.87	516.16	511.45	506.74	449.
826.	536.33	531.62	526.91	522.20	517.49	512.78	508.07	447.
828.	537.65	532.94	528.23	523.52	518.81	514.10	509.39	445.
830.	538.97	534.26	529.55	524.84	520.13	515.42	510.71	443.
832.	540.30	535.59	530.88	526.17	521.46	516.75	512.04	441.
834.	541.62	536.91	532.20	527.49	522.78	518.07	513.36	439.
836.	542.94	538.23	533.52	528.81	524.10	519.40	514.69	437.
838.	544.27	539.56	534.85	530.14	525.43	520.72	516.01	435.
840.	545.59	540.88	536.17	531.46	526.75	522.04	517.33	433.
842.	546.92	542.21	537.50	532.79	528.08	523.37	518.66	431.
844.	548.24	543.53	538.82	534.11	529.40	524.69	519.98	429.
846.	549.56	544.85	540.14	535.43	530.72	526.01	521.30	427.
848.	550.89	546.18	541.47	536.76	532.05	527.34	522.63	425.
850.	552.17	547.50	542.79	538.08	533.37	528.66	523.95	423.
852.	553.29	548.83	544.12	539.41	534.70	529.99	525.28	421.
854.	554.23	550.15	545.44	540.73	536.02	531.31	526.60	419.
856.	554.99	551.47	546.76	542.05	537.34	532.63	527.92	417.
858.	555.59	552.80	548.09	543.38	538.67	533.96	529.25	415.
860.	556.00	554.12	549.41	544.70	539.99	535.28	530.57	413.
862.	556.24	555.44	550.73	546.02	541.31	536.60	531.89	411.
864.	556.31	556.31	552.03	547.33	542.62	537.90	533.19	409.
866.	556.31	556.31	553.17	548.53	543.84	539.14	534.44	407.
868.	556.31	556.31	554.13	549.66	545.03	540.35	535.66	405.
870.	556.31	556.31	554.92	550.70	546.14	541.52	536.84	403.
872.	556.31	556.31	555.53	551.64	547.20	542.63	538.01	401.
874.	556.31	556.31	555.96	552.50	548.21	543.70	539.11	399.
876.	556.31	556.31	556.22	553.28	549.15	544.74	540.20	397.
878.	556.31	556.31	556.30	553.96	550.05	545.72	541.24	395.
880.	556.31	556.31	556.31	554.56	550.87	546.66	542.27	393.
882.	556.31	556.31	556.31	555.08	551.64	547.57	543.23	391.
884.	556.31	556.31	556.31	555.49	552.34	548.41	544.17	389.
886.	556.31	556.31	556.31	555.82	553.00	549.22	545.08	387.
888.	556.31	556.31	556.31	556.08	553.59	550.00	545.96	385.
890.	556.31	556.31	556.31	556.23	554.12	550.71	546.78	383.
892.	556.31	556.31	556.31	556.30	554.60	551.39	547.58	381.
894.	556.31	556.31	556.31	556.31	555.02	552.03	548.35	379.
896.	556.31	556.31	556.31	556.31	555.39	552.62	549.08	377.
898.	556.31	556.31	556.31	556.31	555.68	553.16	549.77	375.

Apprentice engineer (A/E) Duties

Tank sounding (Fuel oil, Sludge oil, Bilge water, e.t.c.)

Quiz 3)

Sounding depth = 861 cm / Trim = By the stern -1.0 m

How much is the volume of tank => (549.41+550.73)/2 = 550.07 m^3

Quiz 4)

Sounding depth = 861 cm / Trim = By the stern -1.5 m

How much is the volume of tank =>

$$\frac{(549.41+544.70)/2 + (550.73+546.02)/2}{2} = 547.715\text{m}^3 \cong 547.72\text{m}^3$$

*** TANK SOUNDING TABLES ***

SOUNDING DEPTH CM	TRIM HEAD 1.0M	EVEN KEEL 0.0M	CAPACITIES IN CUBIC METERS TRIM BY THE STERN -1.0M	-2.0M	-3.0M	-4.0M	-5.0M	ULLAGE DEPTH CM
800.	519.12	514.41	509.70	504.99	500.28	495.57	490.86	473.
802.	520.44	515.73	511.02	506.31	501.60	496.89	492.18	471.
804.	521.76	517.05	512.34	507.63	502.92	498.21	493.50	469.
806.	523.09	518.38	513.67	508.96	504.25	499.54	494.83	467.
808.	524.41	519.70	514.99	510.28	505.57	500.86	496.15	465.
810.	525.73	521.02	516.31	511.60	506.89	502.19	497.48	463.
812.	527.06	522.35	517.64	512.93	508.22	503.51	498.80	461.
814.	528.38	523.67	518.96	514.25	509.54	504.83	500.12	459.
816.	529.71	525.00	520.29	515.58	510.87	506.16	501.45	457.
818.	531.03	526.32	521.61	516.90	512.19	507.48	502.77	455.
820.	532.35	527.64	522.93	518.22	513.51	508.80	504.09	453.
822.	533.68	528.97	524.26	519.55	514.84	510.13	505.42	451.
824.	535.00	530.29	525.58	520.87	516.16	511.45	506.74	449.
826.	536.33	531.62	526.91	522.20	517.49	512.78	508.07	447.
828.	537.65	532.94	528.23	523.52	518.81	514.10	509.39	445.
830.	538.97	534.26	529.55	524.84	520.13	515.42	510.71	443.
832.	540.30	535.59	530.88	526.17	521.46	516.75	512.04	441.
834.	541.62	536.91	532.20	527.49	522.78	518.07	513.36	439.
836.	542.94	538.23	533.52	528.81	524.10	519.40	514.69	437.
838.	544.27	539.56	534.85	530.14	525.43	520.72	516.01	435.
840.	545.59	540.88	536.17	531.46	526.75	522.04	517.33	433.
842.	546.92	542.21	537.50	532.79	528.08	523.37	518.66	431.
844.	548.24	543.53	538.82	534.11	529.40	524.69	519.98	429.
846.	549.56	544.85	540.14	535.43	530.72	526.01	521.30	427.
848.	550.89	546.18	541.47	536.76	532.05	527.34	522.63	425.
850.	552.17	547.50	542.79	538.08	533.37	528.66	523.95	423.
852.	553.29	548.83	544.12	539.41	534.70	529.99	525.28	421.
854.	554.23	550.15	545.44	540.73	536.02	531.31	526.60	419.
856.	554.99	551.47	546.76	542.05	537.34	532.63	527.92	417.
858.	555.59	552.80	548.09	543.38	538.67	533.96	529.25	415.
860.	556.00	554.12	549.41	544.70	539.99	535.28	530.57	413.
862.	556.24	555.44	550.73	546.02	541.31	536.60	531.89	411.
864.	556.31	556.31	552.03	547.33	542.62	537.90	533.19	409.
866.	556.31	556.31	553.17	548.53	543.84	539.14	534.44	407.
868.	556.31	556.31	554.13	549.66	545.03	540.35	535.66	405.
870.	556.31	556.31	554.92	550.70	546.14	541.52	536.84	403.
872.	556.31	556.31	555.53	551.64	547.20	542.63	538.01	401.
874.	556.31	556.31	555.96	552.50	548.21	543.70	539.11	399.
876.	556.31	556.31	556.22	553.28	549.15	544.74	540.20	397.
878.	556.31	556.31	556.30	553.96	550.05	545.72	541.24	395.
880.	556.31	556.31	556.31	554.56	550.87	546.66	542.27	393.
882.	556.31	556.31	556.31	555.08	551.64	547.57	543.23	391.
884.	556.31	556.31	556.31	555.49	552.34	548.41	544.17	389.
886.	556.31	556.31	556.31	555.82	553.00	549.22	545.08	387.
888.	556.31	556.31	556.31	556.08	553.59	550.00	545.96	385.
890.	556.31	556.31	556.31	556.23	554.12	550.71	546.78	383.
892.	556.31	556.31	556.31	556.30	554.60	551.39	547.58	381.
894.	556.31	556.31	556.31	556.31	555.02	552.03	548.35	379.
896.	556.31	556.31	556.31	556.31	555.39	552.62	549.08	377.
898.	556.31	556.31	556.31	556.31	555.68	553.16	549.77	375.

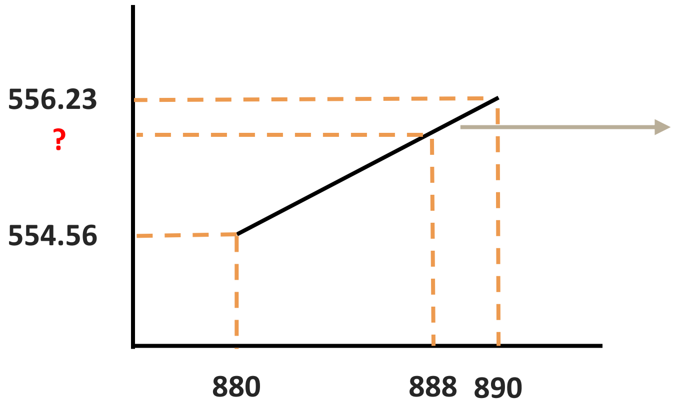
Apprentice engineer (A/E) Duties

Tank sounding (Fuel oil, Sludge oil, Bilge water, e.t.c.)

Quiz 5)

Sounding depth = 888 cm / Trim = By the stern -2.0 m

How much is the volume of tank?



$$Slope = \frac{556.23 - 554.56}{890 - 880} = 0.167$$

$$y = 0.167x + b$$

$$556.23 = 0.167 \times 890 + b$$

$$b = 407.6$$

$$y = 0.167 \times 888 + 407.6$$

$$y = 555.896$$

*** TANK SOUNDING TABLES ***

SOUNDING DEPTH CM	TRIM HEAD 1.0M	EVEN KEEL 0.0M	CAPACITIES IN CUBIC METERS TRIM BY THE STERN -1.0M	-2.0M	-3.0M	-4.0M	-5.0M	ULLAGE DEPTH CM
800.	519.12	514.41	509.70	504.99	500.28	495.57	490.86	473.
810.	525.73	521.02	516.31	511.60	506.89	502.19	497.48	463.
820.	532.35	527.64	522.93	518.22	513.51	508.80	504.09	453.
830.	538.97	534.26	529.55	524.84	520.13	515.42	510.71	443.
840.	545.59	540.88	536.17	531.46	526.75	522.04	517.33	433.
850.	552.17	547.50	542.79	538.08	533.37	528.66	523.95	423.
860.	556.00	554.12	549.41	544.70	539.99	535.28	530.57	413.
870.	556.31	556.31	554.92	550.70	546.14	541.52	536.84	403.
880.	556.31	556.31	556.31	554.56	550.87	546.66	542.27	393.
890.	556.31	556.31	556.31	556.23	554.12	550.71	546.78	383.

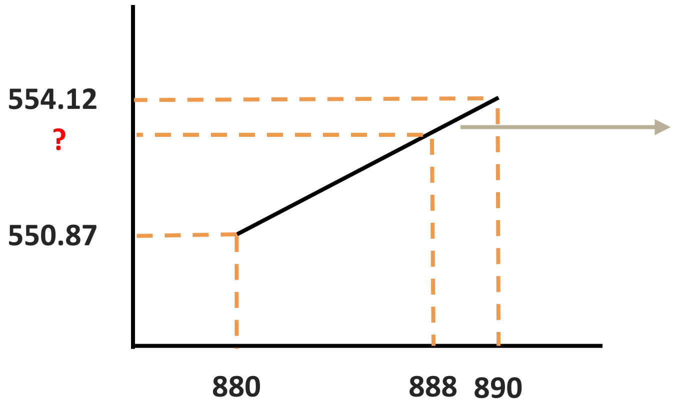
Apprentice engineer (A/E) Duties

Tank sounding (Fuel oil, Sludge oil, Bilge water, e.t.c.)

Quiz 5)

Sounding depth = 888 cm / Trim = By the stern -3.0 m

How much is the volume of tank?



$$Slope = \frac{554.12 - 550.87}{890 - 880} = 0.325$$

$$y = 0.325x + b$$

$$554.12 = 0.325 \times 890 + b$$

$$b = 264.87$$

$$y = 0.325 \times 888 + 264.87$$

$$y = 553.47$$

*** TANK SOUNDING TABLES ***

SOUNDING DEPTH CM	TRIM HEAD 1.0M	EVEN KEEL 0.0M	CAPACITIES IN CUBIC METERS TRIM BY THE STERN -1.0M	-2.0M	-3.0M	-4.0M	-5.0M	ULLAGE DEPTH CM
800.	519.12	514.41	509.70	504.99	500.28	495.57	490.86	473.
810.	525.73	521.02	516.31	511.60	506.89	502.19	497.48	463.
820.	532.35	527.64	522.93	518.22	513.51	508.80	504.09	453.
830.	538.97	534.26	529.55	524.84	520.13	515.42	510.71	443.
840.	545.59	540.88	536.17	531.46	526.75	522.04	517.33	433.
850.	552.17	547.50	542.79	538.08	533.37	528.66	523.95	423.
860.	556.00	554.12	549.41	544.70	539.99	535.28	530.57	413.
870.	556.31	556.31	554.92	550.70	546.14	541.52	536.84	403.
880.	556.31	556.31	556.31	554.56	550.87	546.66	542.27	393.
890.	556.31	556.31	556.31	556.23	554.12	550.71	546.78	383.

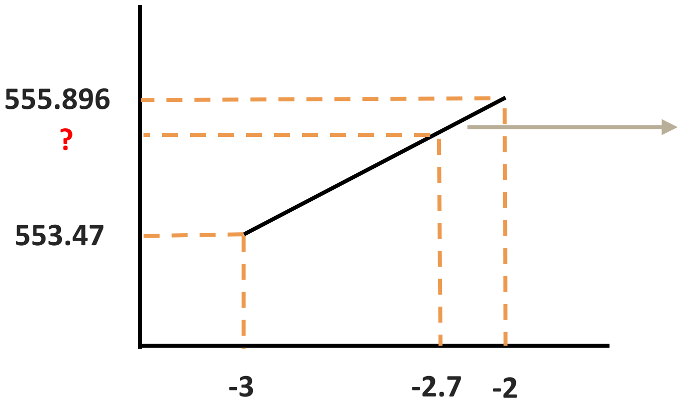
Apprentice engineer (A/E) Duties

Tank sounding (Fuel oil, Sludge oil, Bilge water, e.t.c.)

Quiz 5)

Sounding depth = 888 cm / Trim = By the stern -2.7 m

How much is the volume of tank?



$$Slope = \frac{555.896 - 553.47}{-2-(-3)} = 2.426$$

$$y = 2.426x + b$$

$$553.47 = 2.426 \times (-3) + b$$

$$b = 560.748$$

$$y = 2.426 \times (-2.7) + 560.748$$

$$y = 554.1978$$

*** TANK SOUNDING TABLES ***

SOUNDING DEPTH CM	TRIM HEAD 1.0M	EVEN KEEL 0.0M	CAPACITIES IN CUBIC METERS TRIM BY THE STERN -1.0M	-2.0M	-3.0M	-4.0M	-5.0M	ULLAGE DEPTH CM
800.	519.12	514.41	509.70	504.99	500.28	495.57	490.86	473.
810.	525.73	521.02	516.31	511.60	506.89	502.19	497.48	463.
820.	532.35	527.64	522.93	518.22	513.51	508.80	504.09	453.
830.	538.97	534.26	529.55	524.84	520.13	515.42	510.71	443.
840.	545.59	540.88	536.17	531.46	526.75	522.04	517.33	433.
850.	552.17	547.50	542.79	538.08	533.37	528.66	523.95	423.
860.	556.00	554.12	549.41	544.70	539.99	535.28	530.57	413.
870.	556.31	556.31	554.92	550.70	546.14	541.52	536.84	403.
880.	556.31	556.31	556.31	554.56	550.87	546.66	542.27	393.
890.	556.31	556.31	556.31	556.23	554.12	550.71	546.78	383.

Apprentice engineer (A/E) Duties

Tank sounding (Fuel oil, Sludge oil, Bilge water, e.t.c.)

Apprentice engineer (A/E) Duties

Tank sounding (Fuel oil, Sludge oil, Bilge water, e.t.c.)

Apprentice engineer (A/E) Duties

Tank sounding (Fuel oil, Sludge oil, Bilge water, e.t.c.)

Apprentice engineer (A/E) Duties

Tank sounding (Fuel oil, Sludge oil, Bilge water, e.t.c.)

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Apprentice engineer (A/E) Duties

Tank sounding (Fuel oil, Sludge oil, Bilge water, e.t.c.)

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Apprentice engineer (A/E) Duties

Cooling fresh water test (Please refer to another ppt file) & Boiler water test

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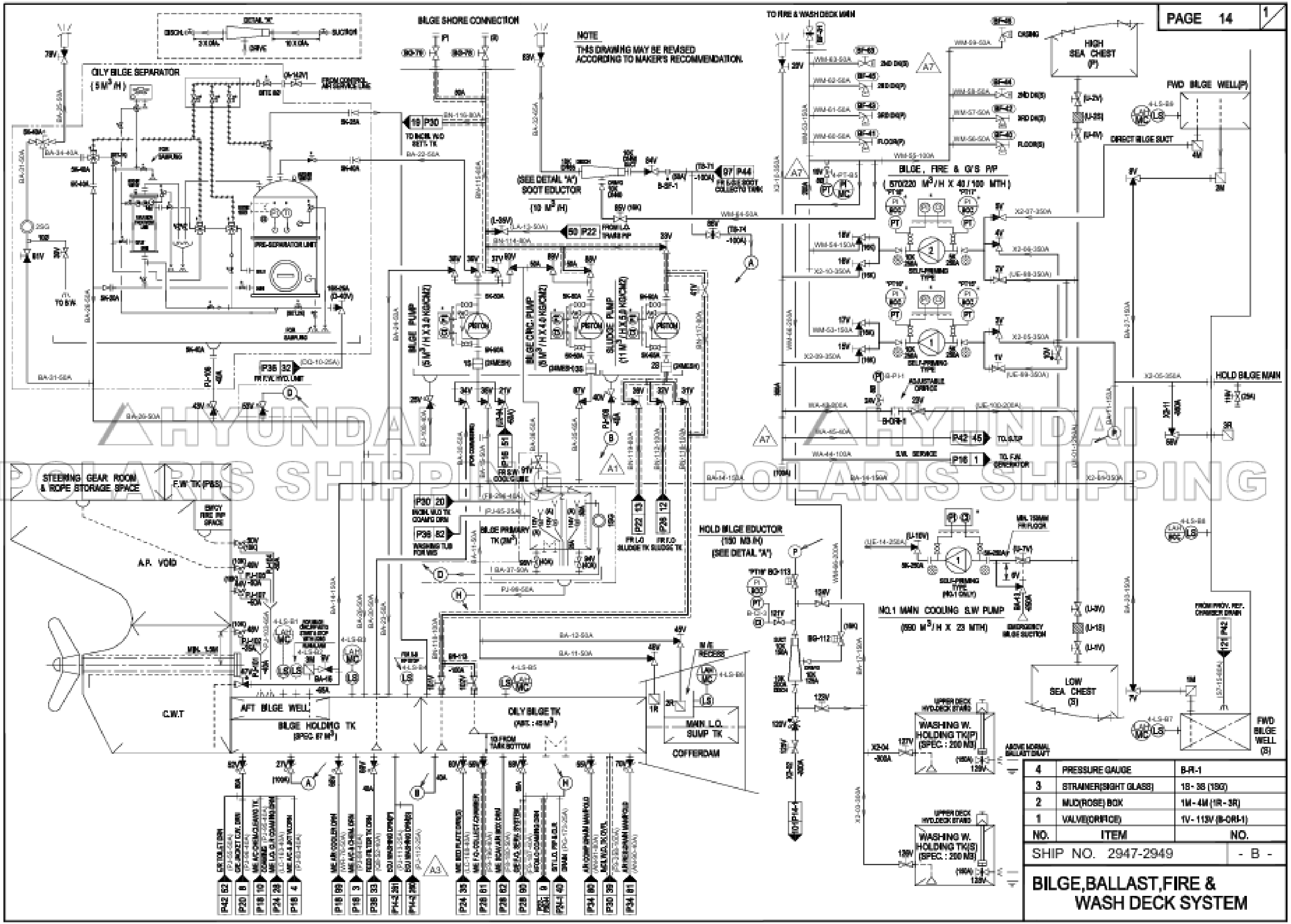
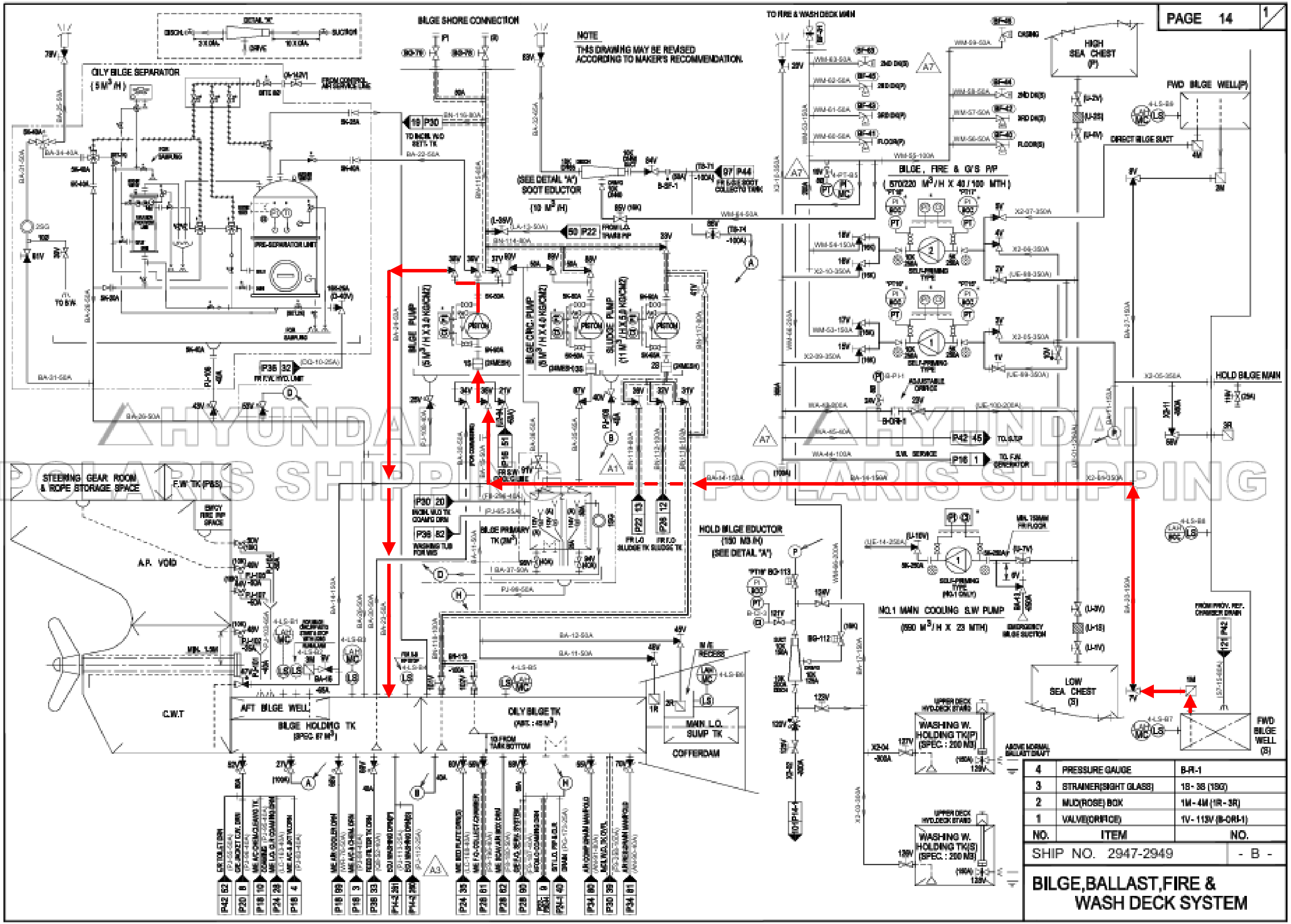
Apprentice engineer (A/E) Duties

Transfer of bilge water from the bilge well to the bilge holding tank


PAGE 14
BILGE SHORE CONNECTION
NOTE
THIS DRAWING MAY BE REVISED ACCORDING TO MAKER'S RECOMMENDATION.
TO FIRE & WASH DECK MAIN
HIGH SEA CHEST (P)
FWD BILGE WELL(P)
DIRECT BILGE SUCT
OILY BILGE SEPARATOR
PRE-SEPARATOR UNIT
SOOT EDUCTOR
FR L.O. SUMP TK
BILGE, FIRE & G/S P/P
SELF-PRIMING TYPE
ADJUSTABLE ORIFICE
HOLD BILGE MAIN
BILGE PUMP
BILGE CIRC. PUMP
SLUDGE PUMP
BILGE PRIMARY TK
STEERING GEAR ROOM & ROPE STORAGE SPACE
F.W. TK (P&S)
EMCY FIRE P/P SPACE
A.P. VOID
C.W.T
AFT BILGE WELL
BILGE HOLDING TK
OILY BILGE TK
MAIN L.O. SUMP TK
COFFERDAM
HOLD BILGE EDUCTOR
(SEE DETAIL "A")
NO.1 MAIN COOLING S.W PUMP
EMERGENCY BILGE SUCTION
LOW SEA CHEST (S)
WASHING W. HOLDING TK(P) (SPEC. : 200 M3)
WASHING W. HOLDING TK(S) (SPEC. : 200 M3)
ABOVE NORMAL BALLAST DRAFT
FWD BILGE WELL (S)
FR F.O. SLUDGE TK
TO S.T.P
TO F.W. GENERATOR
S.W. SERVICE
DETAIL "A"
DISCH.
SUCTION
DRIVE
4 PRESSURE GAUGE B-R-1
3 STRAINER(SIGHT GLASS) 1S - 3S (1SG)
2 MUD(ROSE) BOX 1M - 4M (1R - 3R)
1 VALVE(ORIFICE) 1V - 113V (B-OR-1)
NO. ITEM NO.
SHIP NO. 2947-2949 - B -
BILGE,BALLAST,FIRE & WASH DECK SYSTEM



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Apprentice engineer (A/E) Duties

Transfer of bilge water from the bilge well to the bilge holding tank

Apprentice engineer (A/E) Duties

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Apprentice engineer (A/E) preparation materials

You can prepare the materials by referring to the Excel file uploaded to GitHub and add any additional items you need.

승선 시 준비물

BUY: 아직 준비가 되지 않아 구매가 필요한 것
CHECK: 준비가 완료되어 캐리어 안에 집어 넣은 것

	NAME	Q'TY	REMARK	BUY	CHECK
1	근무복	1	하복/동복 상하의, 벨트		
2	체육복	1	봄가을용, 여름용		
3	필기구	N	필통, 펜, 샤프, 화이트, 형광펜		
4	LED 손전등	1	손전등은 한 손에 잡히고 주머니에 넣을 수 있는 크기. 배터리, 충전기, 충전 케이블		
5	전자손목시계	1	Flowmeter를 통한 기름 계산할 때 필요함. 초시계 잴 수 있는 것. "카시오 F-91W-1D" 추천		
6	외장하드	1	4TB 정도면 충분		
7	USB	1			
8	개인서적	N			
9	실습 리포트 준비물	N			
10	반팔	6	일상용 : 4개 , 상륙용 : 2개		
11	바지	4	반바지 1개, 일반바지 2개		
12	수건	5			
13	세면도구	1	치약 1개, 칫솔 2개, 치간칫솔 n개		
14	선글라스	1			
15	양말	10	검은색 : 2개 , 일반 : 8개		
16	운동화	1	승선할 때 신고 가면 됨		
17	손톱깎이	1			
18	전기면도기	1	충전기 같이 챙기기		
19	의약품	N	본인에게 꼭 필요한 것만 챙기기		
20	노트북	1	전원 선, 마우스, 노트북 가방		
21	핸드폰	1	충전기 같이 챙기기		
22	이어폰	1			
23	립크루즈	1			
24	손 수첩	1	기관실에서 실습 시 메모용 (주머니에 들어갈 사이즈) "양지사 피디수첩" 추천		
25	필기 노트	1			
26	지갑	1			
27	속옷	8	일반용 : 5, 작업복 입을 시 : 3		
28	안대	1			
29	로션	1			

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Future career direction

First of all, what I'm writing here isn't the definitive answer to your career path. There are certainly better options out there, of which I'm unaware. So, please use the following information as a reference only.

Students majoring in marine systems engineering at KMOU are divided into four major groups after graduation.

1) Onboard the ship (Majorly for military service)

2) Military enlistment (ROTC)

3) Get a job at a company (Women and men who served in the military)

4) Graduate school admission (Women and men who served in the military)

Future career direction

1) Onboard the ship (Majorly for military service)

If you are boarding a ship, it is recommended to try to become a 1st engineer if possible.

The condition for obtaining a 1st class engineer license is to board a ship for one year as a 1st engineer while holding a 2nd class engineer license.

Therefore, you can meet the requirements by boarding a ship for about 5 to 6 years.

However, if given the opportunity, you can challenge yourself to become a chief engineer.

These days, promotions are quick, so there are cases where you can be promoted to chief engineer in your early 30s.

Future career direction

1) Onboard the ship (Majorly for military service)

If you have experience as a marine engineer, you can challenge yourself for the following jobs.

- Shipping companies (Where you worked or elsewhere)
- Shipbuilding companies
- Shipbuilding and marine equipment companies & Others
- Classification societies & Flag registry
- Ministry of Oceans and Fisheries
- Coast Guard
- Public institutions and corporations

Future career direction

1) Onboard the ship (Majorly for military service)

If you have experience as a marine engineer, you can challenge yourself for the following jobs.

- Shipping companies (Where you worked or elsewhere)

Future career direction

1) Onboard the ship (Majorly for military service)

If you have experience as a marine engineer, you can challenge yourself for the following jobs.

- Shipbuilding companies

HD한국조선해양
HD현대중공업
HD현대삼호
HD현대마린솔루션
HD현대일렉트릭
HD현대사이트솔루션
HD현대인프라코어
SAMSUNG 삼성중공업
HJ 중공업
한화오션
한화엔진 Hanwha Engine
케이조선
HSG 성동조선

국립한국해양대학교
NATIONAL KOREA MARITIME & OCEAN UNIVERSITY

Future career direction

1) Onboard the ship (Majorly for military service)

Shipbuilding and marine equipment companies & Others

Future career direction

1) Onboard the ship (Majorly for military service)

If you have experience as a marine engineer, you can challenge yourself for the following jobs.

- Classification societies & Flag registry

Future career direction

1) Onboard the ship (Majorly for military service)

If you have experience as a marine engineer, you can challenge yourself for the following jobs.

- Ministry of Oceans and Fisheries 해양수산부

해양수산부 동해어업관리단

해양수산부 부산지방해양수산청

해양수산부 울산지방해양수산청

해양수산부 서해어업관리단

국립해양조사원

해양수산부 포항지방해양수산청

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해양수산부 여수지방해양수산청

국립한국해양대학교 NATIONAL KOREA MARITIME & OCEAN UNIVERSITY

Future career direction

1) Onboard the ship (Majorly for military service)

If you have experience as a marine engineer,

you can challenge yourself for the following jobs.

- Coast Guard

Future career direction

1) Onboard the ship (Majorly for military service)

If you have experience as a marine engineer, you can challenge yourself for the following jobs.

- Public institutions and corporations

한국가스공사
KOREA GAS CORPORATION

한국전력공사
KOREA ELECTRIC POWER CORPORATION

한국지역난방공사
KOREA DISTRICT HEATING CORP.

한국가스기술공사
KOREA GAS TECHNOLOGY CORPORATION

E&C 한국전력기술
KEPCO ENGINEERING & CONSTRUCTION

KNOC
한국석유공사
KOREA NATIONAL OIL CORPORATION

한국가스안전공사
KOREA GAS SAFETY CORPORATION

EWP
한국동서발전(주)
KOREA EAST-WEST POWER CO.,LTD.

KOMIPO
한국중부발전

wp
한국서부발전(주)

한국수력원자력(주)
KOREA HYDRO & NUCLEAR POWER CO., LTD

KOEN
한국남동발전

한국남부발전(주)

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Future career direction

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마이스터고등학교 완도수산고등학교

Future career direction

2) Military enlistment (ROTC)

After ROTC, you can develop your expertise by boarding a ship in shipping company, and consider other paths such as employment at a shipbuilding companies or the Coast Guard, or going on to graduate school.

If you board a ship after ROTC, the company will generally promote you quickly.

Long-term service in the military is also an option.

Because society sometimes gives preference to ROTC graduates, you can apply to various companies in addition to the maritime companies mentioned above.

Future career direction

3) Get a job at a company (Women and men who served in the military)

If you apply for a job right after graduating from university, you can apply to a variety of fields just like students at regular universities.

However, applying to a company related to marine engineering is very advantageous compared to other general university students.

Companies that put up job-related placards and posters at university can be considered to be favorable to the university.

When applying to companies in different fields than ours, you will need to prepare specifications that are appropriate for the position.

Future career direction

4) Graduate school admission

If finding a job is difficult, graduate school may be an alternative option.

However, I personally recommend that you look for a job after graduating with a master's or doctorate degree rather than looking for a job while attending graduate school.

Going to graduate school after boarding the ship is also a good career path.

The best reason to apply for graduate school is to have a clear sense of purpose about your career path after graduate school. Because, during graduate school, you have to sit in a chair and focus on research most of the time, so it is a different environment from usual, and you also have to enjoy research, have many new ideas, and have a challenging attitude.

Future career direction

4) Graduate school admission

Currently, the competition rate for admission to KMOU's graduate school is somewhat high.

This is because a high percentage of working students (=currently working at company) choose to pursue graduate school part-time. However, if you pursue graduate school full-time, your chances of admission significantly increase.

When choosing a graduate school, it's important to confirm whether the professor's research area aligns with your desired direction. This can be determined by reviewing the papers listed on the professor's website. (Carefully check the number and quality of papers written by the professor as the first author. Identify the first author rather than co-authors and corresponding authors.)

Even if it's not KMOU, there are many excellent graduate schools in Korea and abroad, so it's not a bad idea to think about it from various perspectives.

Future career direction

4) Graduate school admission

Below are some research positions that I believe you might be interested in pursuing with a marine systems engineering degree. Please note that not all positions are open to you; only positions that align with your major are eligible.

KRISO 선박해양플랜트연구소

국방과학연구소 Agency for Defense Development

국가정보원 NATIONAL INTELLIGENCE SERVICE

KITECH 한국생산기술연구원

KMC 한국해사협력센터 Korea Maritime Cooperation Center

KOMERI 한국조선해양기자재연구원 Korea Marine Equipment Research Institute

KIOST 한국해양과학기술원

KMI 한국해양수산개발원 KOREA MARITIME INSTITUTE

RIMS 중소조선연구원

ETRI 한국전자통신연구원 Electronics and Telecommunications Research Institute

KOMSA 한국해양교통안전공단 KOREA MARITIME TRANSPORTATION SAFETY AUTHORITY

KR KOREAN REGISTER

KIMM 한국기계연구원 KOREA INSTITUTE OF MACHINERY & MATERIALS

KERI 한국전기연구원 KOREA ELECTROTECHNOLOGY RESEARCH INSTITUTE

KIER 한국에너지기술연구원 KOREA INSTITUTE OF ENERGY RESEARCH

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Future career direction

A challenge that takes time but is worth it

1) Coast Guard officer candidate

2) Maritime law schools (Korea University, Jeju National University, etc.)

3) National Intelligence Service, Grade 7

4) Ministry of Oceans and Fisheries, Grade 5

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